

A Choice Model of Gradual Attention ^{*}

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Abstract

Despite a growing interest in choice models with limited attention, there is a lack of studies modeling limited attention itself. In this paper, I propose a choice model with gradual attention, in which a decision maker's attention is limited due to time constraints. With richer choice data including the associated decision time, I provide an axiomatic characterization of this model. I show the model primitives such as the preference can be uniquely identified. To make the model comparable to existing choice models with limited attention, I consider its static version without time data and show identification is not always possible, which highlights the novelty of utilizing the decision time information in understanding choice behaviors. I examine aggregation issues of this model and give necessary and sufficient conditions for the existence of a representative agent under heterogeneity. In my model, the decision maker's consideration set is growing monotonically as decision time increases. As a consequence, more time leads to better choices. I test this choice pattern using experimental data from a well-known study and find supportive evidence.

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1 Introduction

The classic revealed preference theory assumes a decision maker (DM) considers all available options and selects the best one based on a preference relation. A large number of empirical studies find evidence against the classic revealed preference theory. It is widely believed the DM’s rationality can be constrained by various behavioral limitations. A growing literature attributes choice anomalies to limited attention, that is, the DM’s consideration set is a subset of all available options. The DM is assumed to choose the most preferred option from the consideration set. Early works of limited attention models focus on deterministic choices, where consideration sets are deterministic (see Masatlioglu, Nakajima, and Ozbay (2012), Lleras et al. (2017), and Dean, Kıbrıs, and Masatlioglu (2017)). A few recent papers extend limited attention models to study stochastic choices (see Manzini and Mariotti (2014) and Brady and Rehbeck (2016)), in which consideration sets are random. In these random consideration models, the DM’s preference is deterministic and the randomness in choices comes exclusively from the randomness of consideration sets.

Although choice models with limited attention are popular in economics, there is a lack of studies modeling limited attention itself. Most theoretical models of limited attention assume specific forms of consideration sets, without providing a justification of why the DM’s attention is limited in the first place. In this paper, I study a choice model with gradual attention, where the DM’s attention is limited due to time constraints. The deterministic rational choice model is a limiting case of my model, when time constraints are not binding. The idea of modeling limited attention as a consequence of limited time is naturally motivated by the fact that in reality, evaluating options, such as hundreds of mobile phones from different brands, can be extremely time-consuming. Given time constraints, it is infeasible for the DM to evaluate every single mobile phone.

In the gradual attention model, the DM evaluates a menu of options sequentially and her consideration set consists of only those evaluated options. She then chooses her favorite option from the consideration set, based on her deterministic preference. Her consideration is not necessarily the entire menu because she may not have enough time to finish evaluating all available options. The DM’s order of evaluating options is called “salience order”. The salience order measures the salience of the options and one example is the list order ¹ in online shopping platforms like Amazon. It is common for a consumer to sort the list based on certain criteria and to evaluate items following the list order. The salience order may be purely subjective and is difficult to be directly observed. Reutskaja et al. (2011) uses

¹See Rubinstein and Salant (2006), Rubinstein and Salant (2011), and Yildiz (2016) for models with lists.

eye-tracking data to monitor the search process of subjects and find subjects search more often in certain focal regions. Options in these focal regions can be interpreted as the most salient options. The DM’s evaluation process is modeled by a generic counting process. The possible randomness in the evaluation process can be flexibly used to model stochastic choice since consideration sets can be stochastic.

I provide an axiomatic characterization of the gradual attention model with richer choice data than standard choice models. In standard choice models, it is assumed the DM’s choices (or choice probabilities) from different menus are observed. In my model, I further assume the contemplation time associated with each choice is also observed ². The primitives of the model are a salience order, a counting process, and a preference relation. I show, with the richer data, the DM’s salience order and preference can be uniquely identified. To outline the intuition for identification, consider how the DM’s choice from a doubleton menu $\{a, b\}$ changes as her contemplation time increases. With more time, it is more likely her consideration becomes the entire menu. It follows her choice converges to her most preferred option over time. If the choice probability of a goes up as time increases, it must be the case that b is evaluated before a while the DM prefers a to b .

The consideration set in my model is monotonically growing as time increases. This is conceptually similar with those search-based choice models with limited attention, where options enter the consideration set sequentially (see Masatlioglu and Nakajima (2013), Caplin and Dean (2011), and Caplin, Dean, and Martin (2011)). Since the DM’s preference is deterministic, her choice must be better if her consideration set becomes larger. This further implies the DM makes better choices as time increases and this choice pattern is called “increasing quality”. However, this choice pattern is not shared by all search-based choice models. In search models, because of searching costs, the DM uses certain optimal stopping rules to terminate searching and makes a choice. The relation between the choice quality and the contemplation time typically depends on the specific optimal stopping rule and other details of the search model. Consider the famous satisficing searching model by Simon (1955), where the value of each option is independently drawn from a probability distribution and the DM stops searching if the highest value of searched options exceeds a certain threshold. In Simon (1955)’s model, if the DM is further constrained by a predetermined stopping time, then the choice quality must be positively correlated with time. For a negative correlation, Fudenberg, Strack, and Strzalecki (2018) studies a sequential sampling model with a binary choice problem, where the DM pays a constant time cost to learn utility values of the choices. They show choices are more likely to be correct if decisions are made quickly.

²For example, the contemplation time can be measured by the amount of time a consumer spends on Yelp to choose a restaurant.

Whether the decision time is exogenous or is determined by endogenous optimal stopping rules should depend on application contexts. In many environments, such as grocery shopping and restaurant searching, it is reasonable to believe the time constraints are pre-determined or exogenous. Moreover, to figure out an optimal stopping rule, the DM is often required to have a prior belief about the distribution of the options' utility values. This imposes additional structures on the DM's knowledge and utility function. In contrast, my model only assumes the DM has an ordinal preference. To test my model empirically, I revisit the experiments by Caplin, Dean, and Martin (2011), where subjects have to make choices given time constraints that are exogenously determined. The design of the experiments allows me to identify a causal relationship between the contemplation time and the choice quality. I use the experimental data to test the choice pattern of increasing quality and find supportive evidence.

Since my model assumes richer data than standard choice models, it can be viewed as a generalization of many existing (stochastic) choice models with limited attention. It is thus interesting to study the static version of my model, which does not have the decision time information. On one hand, I show uniquely recovering the preference and the salience order is not always possible. The impossibility result highlights the novelty of making use of choice variations over time as a primary source of identification. On the other hand, I show if the number of universally available options is sufficiently large, identification of the preference and the salience order is almost surely achievable. It follows the decision time information can be viewed as a substitute for the abundance of options, in terms of identifying model primitives. The static version of my model is directly comparable to many existing limited attention models. It turns out to be a special case of the monotone random attention model by Cattaneo et al. (2020) so their analysis can be directly applied here.

So far the choice data in my model is assumed to be generated by a single DM. In the literature of random choice, it is often useful to model stochastic choice data as aggregated choices from a population of heterogeneous agents. Suppose there is a population of heterogeneous agents and each agent's choice behavior is described by my model. A natural question that arises is whether there exists a representative agent whose choices are the aggregated choices from the population of agents. That is, whether the choice model in this paper is closed under aggregation. Studies on the existence of a representative agent in the literature of random choice are still scarce. Apesteguia and Ballester (2016) shows a representative agent exists for a generic random utility model but does not exist for other popular special cases such as the Luce model. I am not aware of any similar study for choice models with limited attention. I examine the existence of a representative agent under heterogeneity for each of the three primitives in my model. I show a representative agent exists under

heterogeneity in counting processes, but not under heterogeneity in preferences or salience orders. I characterize necessary and sufficient conditions for a representative agent to exist under heterogeneity in preferences and salience orders. In a special case when the choice data comes from a single menu, I show a representative agent always exists. It follows within a menu, the choice pattern of increasing quality is robust to aggregation under heterogeneity. The robustness allows me to test the increasing choice quality implication using aggregated choice data in the section of experimental evidence.

2 Gradual Attention Model

2.1 Notations and Assumptions

Let X be a finite set of items. A menu A is a non-empty subset of X and $\mathcal{X} = 2^X \setminus \emptyset$ denotes the set of menus. Let $\mathcal{S}$ denote the set of strict preferences over X , that is, the set of binary relations that are complete, transitive and anti-symmetric. A generic preference is denoted by $\succ \in \mathcal{S}$. Given a preference $\succ \in \mathcal{S}$, its induced choice function $\mathcal{C}_\succ : \mathcal{X} \rightarrow X$ is defined as $\mathcal{C}_\succ(A) = \{a \in A : a \succ b \text{ for } \forall b \in A\}$.

Let $t \in \mathcal{T}$ be the amount of time used by the DM to make a decision, where $\mathcal{T}$ is the time index set. A random choice rule with time $\rho : X \times \mathcal{X} \times \mathcal{T} \rightarrow [0, 1]$ satisfies $\sum_{a \in A} \rho(a, A, t) = 1$ for all $t \in \mathcal{T}$. The amount of time t can be interpreted as the deadline of the choice problem so $\rho(a, A, t)$ is the probability that a being chosen from the menu A , given the deadline t . The time index $\mathcal{T}$ can either be discrete or continuous. Given a menu A , I denote the support of A as the set of items that are chosen with non-zero probabilities at some time $s \in \mathcal{T}$, that is, $\text{supp}(A) := \{a \in A : \rho(a, A, s) > 0 \text{ for some } s \in \mathcal{T}\}$.

Let $\{N_t\}_{t \in \mathcal{T}}$ (or N for convenience) be a counting process. For any time $t \in \mathcal{T}$, let μ_t^N be its induced probability distribution over $\mathbb{Z}^+$, that is, $\mu_t^N(Z) := \Pr(N_t = Z)$. A counting process is used to model the size of the consideration set. Throughout the paper, counting processes are restricted to satisfy the following two properties: 1. $\mu_t^N(0) = 0$ for any $t \in \mathcal{T}$ ³; 2. for each $1 \leq Z \leq |X|$, there exists some $t \in \mathcal{T}$ such that $\mu_t^N(Z) > 0$. The second property is satisfied by most commonly used counting processes such as a Poisson process. The set of counting processes satisfying these two properties is denoted by $\mathcal{N}$.

A generic salience order is denoted by $R \in \mathcal{S}$ and aRb implies a is evaluated before b . Given a positive integer $Z \in \mathbb{Z}^{++}$, a menu $A \in \mathcal{X}$ and a salience order $R \in \mathcal{S}$, the R induced Z -consideration set of menu A is $A^R(Z) := \{a \in A : |\{b \in A : bRa\}| < \min\{Z, |A|\}\}$. In other words, $A^R(Z)$ contains the Z most salient items of A , if $Z < |A|$. If $Z \geq |A|$, then

³This restriction guarantees the consideration set is non-empty.

$$A^R(Z) = A.$$

Definition 1. (Gradual Attention Representation)

A random choice rule with time ρ is said to have a gradual attention representation $(\succ, R, N)$ if for all $a \in X, A \in \mathcal{X}, t \in \mathcal{T}$,

$$\rho(a, A, t) = \sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ}(A^R(Z)) = a\}} \mu_t^N(Z) \quad (1)$$

The gradual attention representation describes the following decision making process: the DM has a true preference $\succ$ and is equipped with a salience order R . She needs to make a choice from a menu A by the deadline t . She may not be able to pay attention to all items in A due to time constraint. The items enter her consideration sequentially, following the salience order R . The size of her consideration set depends on the the realization of the counting process N at time t . She then chooses the best item from her consideration set based on her preference $\succ$. The DM's consideration set formation process can be alternatively interpreted as a process of learning or searching. The DM learns the utility value of items sequentially, from the most salient one to the least salient one.

Standard choice models assume the DM's choices from different menus are observed, while the gradual attention model further assumes the decision time associated with each choice is also observed. To exploit the information of decision time, I assume there are sufficient choice variations over time.

Definition 2. (Time Variation Assumption)

A random choice rule with time ρ is said to satisfy the time variation assumption if for any menu $A \in \mathcal{X}, a \in A$, $|\rho(a, A, \mathcal{T})| = 1 \iff \rho(a, A, \mathcal{T}) \subset \{0, 1\}$, where $\rho(a, A, \mathcal{T}) = \{\rho(a, A, t) : t \in \mathcal{T}\}$.

The time variation assumption requires the choice probability of any item to vary over time unless the item is either always chosen or never chosen. The time variation assumption fails if a static version of the gradual attention model is considered and there is no choice variation over time. There are various identification issues in the static version of the model and I will discuss them later.

2.2 Identification

Before providing an axiomatic characterization of the gradual attention model, it is useful to discuss identifying the model primitives. It turns out identification of $\succ, R$ can be achieved from choice variations over time in doubleton menus.

For any $t \in \mathcal{T}$, let $\Delta_{ab}(t) := \rho(a, \{a, b\}, t) - \rho(b, \{a, b\}, t)$. a is said to *dominate* b if $\Delta_{ab}(t) = 1$ (or equivalently $\Delta_{ba}(t) = -1$) for all $t \in \mathcal{T}$. a is said to *impact* b if $\Delta_{ab}(t)$ is weakly increasing in t and is not constant, which is denoted by $\Delta_{ab}(t) \uparrow$ (or equivalently $\Delta_{ba}(t) \downarrow$). Suppose ρ has a gradual attention representation (1). If $a \succ b$, then b is never chosen except when the consideration set contains only b . This requires bRa , since otherwise the consideration set always contains a . As time increases, the probability that the consideration set becomes $\{a, b\}$ goes up and so does the choice probability of a . The above reasoning is summarized as the following proposition.

Proposition 1. (*Identification from Doubletons*)

Suppose ρ satisfies the time variation assumption and has a gradual attention representation $(\succ, R, N)$. Then for any $a, b \in X$, $\Delta_{ab}(t)$ is monotonic in t and is constant if and only if $\Delta_{ab}(t) = 1$ or -1 . Moreover,

- a dominates b , that is, $\Delta_{ab}(t) = 1 \iff a \succ b, aRb$
- b dominates a , that is, $\Delta_{ab}(t) = -1 \iff b \succ a, bRa$
- a impacts b , that is, $\Delta_{ab}(t) \uparrow \iff a \succ b, bRa$
- b impacts a , that is, $\Delta_{ab}(t) \downarrow \iff b \succ a, aRb$

It follows that when $|X| = 2$, ρ has a gradual attention representation with a unique pair of $\succ, R$ if and only if $\Delta_{ab}(t)$ is monotonic in t .

The Proposition 1 shows how to identify $\succ, R$ over an arbitrary pair of items from their doubleton menus. When $|X| = 2$, the monotonicity of $\Delta(t)$ alone characterizes the gradual attention model. When $|X|$ contains more than two items, the monotone choice pattern needs to be generalized to describe monotonic movements of choice probability distributions. The canonical first-order stochastic dominance (FOSD) is introduced to compare two choice probability distributions.

Definition 3. (First-order Stochastic Dominance)

Let A be a set that is ordered by $P \in \mathcal{S}$. Let ω, ν be two probability distribution over A . ν is said to first-order stochastically dominates ω with respect to P , denoted by $\omega \succ_F \nu$ w.r.t $>_P$, if for any $a \in A$,

$$\sum_{\{b \in A: aPb\}} \omega(b) \leq \sum_{\{b \in P: aPb\}} \nu(b)$$

Proposition 2. *(Increasing Choice Quality)*

Suppose ρ has a gradual attention representation $(\succ, R, N)$. Then for any menu A , $\rho(\cdot, A, t) \succ_F \rho(\cdot, A, s)$ w.r.t $\succ$ for all $t \geq s$.

Proposition 2 generalizes the monotone choice pattern in doubleton menus. Consider an expected utility maximizer whose preference over deterministic outcomes is given by $\succ$. It is a well known result that $\omega \succ_F \nu$ w.r.t $\succ$ if and only if the expected utility of ω is higher than that of ν . It follows that Proposition 2 can be interpreted as the DM will make better choices with higher expected utility as time increases. It is natural to expect if the DM has sufficient time to make a decision, she should choose the $\succ$ -best item. The following result confirms this intuition, under a mild assumption on the counting process. This assumption guarantees the consideration set converges to the entire menu and is satisfied by most commonly used counting processes such as a Poisson process.

Corollary 1. *(Gradual Rational Choice)*

Assume for any $\epsilon > 0$, $K \in \mathbb{Z}^{++}$, there exists $s \in \mathcal{T}$ such that $Pr(N_s < K) = \mu_s(< K) < \epsilon$ ⁴. Then ρ converges to $\mathcal{C}_\succ$ in probability as $t \rightarrow \infty$.

Corollary 1 shows the deterministic rational choice model is a limiting case of the gradual attention model. Moreover, Proposition 2 says the convergence is monotonic in the sense that the expected utility loss due to bounded rationality monotonically vanishes over time.

2.3 Axiomatic Characterization

For any $a \in X$, I denote the set of items that impacts a or dominates a as $\overline{M}(a) := \{b \in X : \Delta_{ab}(t) \downarrow \text{ or } \Delta_{ab}(t) = -1\}$. The set of items that dominate a or are impacted by a is denoted as $\overline{H}(a) := \{b \in X : \Delta_{ab}(t) \uparrow \text{ or } \Delta_{ab}(t) = -1\}$. Proposition 1 implies that b is revealed to be preferred to a if it dominates a or impacts a , and that b is more salient than a if it dominates a or is impacted by a . It follows $\overline{M}(a)$ is the upper contour set of a w.r.t $\succ$ and $\overline{H}(a)$ is the upper contour set of a w.r.t R .

Axiom 1. (Monotone Choices from Doubleton Menus)

For any $a, b \in X$, $\Delta_{ab}(t)$ is monotonic in t .

Axiom 2. (Transitivity of Dominance)

If a dominates b and b dominates c , then a dominates c .

Axiom 3. (Transitivity of Impact)

If a impacts b and b impacts c , then a impacts c .

⁴ $\mu(< K)$ denotes the probability that the number of counts is less than K .

Axiom 1 says choices from doubleton menus are monotonic in time, which is necessary for the representation by Proposition 1. Axiom 2 and 3 require the dominance and impact relations to be transitive. It turns out under Axiom 1, transitivity across impact and dominance relations are implied by Axiom 2 and 3.

Lemma 1. (*Transitivity across Impact and Dominance*)

Assume Axioms 1-3 hold. If a impacts b and b dominates c , then a impacts c or a dominates c ; if a dominates b and b impacts c , then a impacts c or a dominates c .

Axioms 1-3 impose restrictions on ρ only for doubleton menus. The remaining axioms build connections between doubleton menus and larger menus.

Axiom 4. (Independence of Dominance)

For any menu A , $\rho(a, A, t) = 0$ for all $t \in \mathcal{T}$ if and only if there exists $b \in A$ such that b dominates a .

Axiom 4 extends the dominance relation to arbitrary menus. Axiom 4 is also implied by the classic deterministic rational choice model and random utility models if the time component is ignored.

Axiom 5. (Increasing Choice Quality)

Let menu A and $t \geq s$ be arbitrary. For any $a \in A$, $\sum_{c \in \overline{M}(a) \cap A} \rho(c, A, t) \geq \sum_{c \in \overline{M}(a) \cap A} \rho(c, A, s)$.

Axiom 5 generalizes the monotone choice pattern in Axiom 1 from doubleton menus to arbitrary menus, which requires choice probabilities to move monotonically in terms of FOSD w.r.t the revealed preference. Although Axiom 1 seems to be redundant under Axiom 5, it is not implied by Axiom 5. For example, suppose $X = \{a, b\}$ and $\Delta_{ab}(t)$ is not monotonic. Then Axiom 1 does not hold. But $\overline{M}(a)$ and $\overline{M}(b)$ are both empty so Axiom 5 is vacuously true.

Theorem 1. (*Gradual Attention Representation*)

Under the assumption of time variation, ρ has a gradual attention representation $(\succ, R, N)$, where $\succ, R$ are menu independent and unique, if and only if ρ satisfies Axioms 1-5.

The only if part of Theorem 1 comes immediately from Proposition 1 and Proposition 2. For the if part, I sketch the proof here and more details are in the appendix. Axiom 1 allows me to construct a candidate pair of $\succ, R$ from ρ on doubleton menus, following the discussion of identification in Proposition 1. Axiom 2 and Axiom 3 are used to prove the constructed $\succ, R \in \mathcal{S}$. The constructed $\succ, R$ impose a restriction on the support of ρ . In particular, an item a is never being chosen if and only if there exists another item b such that

$b \succ a, bRa$, which is implied by Axiom 4. Given $\succ, R$, choice probabilities can be converted to probabilities over $\mathbb{Z}^{++}$. Axiom 5 implies these converted probabilities over $\mathbb{Z}^{++}$ can be ordered based on FOSD w.r.t the natural numbers. By Lemma 6, there exists a counting process N that induces the probabilities over $\mathbb{Z}^{++}$. The menu independence of $\succ, R$ comes from the construction and the uniqueness follows from Proposition 1.

The counting process N in the gradual attention model is not always menu independent and uniquely identified. It can be easily checked that two different counting processes can induce the same probability distribution μ over $\mathbb{Z}^{++}$. Since the choice rule ρ in my model only depends on μ , I focus on examining the menu independence and uniqueness of μ .

Proposition 3. (*Identification of μ*)

Suppose ρ has a gradual representation with menu independent $\succ, R, N$, then the μ induced by N is unique up to $\max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\overline{H}(a) \cap A|$. In particular, $\max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\overline{H}(a) \cap A| = |X| - 1$ if and only if $\text{supp}(B) = B$ for any menu B .

Proposition 3 shows μ is uniquely identified up to an integer that is generally smaller than $|X|$ and the greatest identification occurs when every menu has a full support. To characterize the gradual attention representation with a menu independent N and its induced μ , Axiom 6 is added.

Axiom 6. (Independence of Inferior Choices)

Let $t \in \mathcal{T}$ and two menus A, B be arbitrary. For any $a \in \text{supp}(A), b \in \text{supp}(B)$, $|\overline{H}(a) \cap A| \geq |\overline{H}(b) \cap B| \iff \sum_{c \in \overline{H}(a) \cap A} \rho(c, A, t) \geq \sum_{d \in \overline{H}(b) \cap B} \rho(d, B, t)$.

The set $\overline{H}(a) \cap A$ contains exactly those items in A that are more salient than a . Axiom 6 says for any menu A and any item $a \in A$, the probability of choosing a more salient item than a is independent of a, A , but only depends on the number of items in A that are more salient than a .

Proposition 4. (*Menu Independence of μ*)

Under the assumption of time variation, ρ has a gradual attention representation with menu independent $(\succ, R, N)$ if and only if ρ satisfies Axioms 1-6. Moreover, $\succ, R$ is unique and the μ induced by the counting process N is unique up to $\max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\overline{H}(a) \cap A|$.

3 Comparative Statics

In this section, I present comparative statics results for the gradual attention model in addition to Proposition 2. Since the counting process N is measuring the DM's evaluation speed, a faster counting process leads to better choices.

Proposition 5. (*Comparative Speed and Choice Quality*)

Let two DMs have the same $\succ, R$ and different counting processes N^1, N^2 . Suppose first DM is evaluating more slowly, that is, $\mu_t^{N^1} \succ_F \mu_t^{N^2}$ w.r.t the natural order for any t . Then for any time t and any menu A , the second DM is making better choices, that is, $\rho_1(\cdot, A, t) \succ_F \rho_2(\cdot, A, t)$ w.r.t $\succ$.

The proof of Proposition 5 is similar with that of Proposition 2 and is omitted. In many applications, the DM's evaluation/searching speed is modeled by a Poisson process with λ and it can be verified that a smaller value of λ is indeed equivalent to the definition of being slower in Proposition 5. It follows the expected utility of a DM goes up if her evaluation speed λ increases.

In the gradual attention model, the DM chooses inferior items because they are more salient. If better items becomes more salient, it is more likely the DM's consideration set contains items of higher quality. In other words, if the salience order R is more aligned with the preference $\succ$, then the DM makes better choices.

Definition 4. (*Comparative Alignments with Preference*)

Let $R_1, R_2, \succ \in \mathcal{S}$. R_1 is said to be *more aligned with $\succ$* than R_2 , denoted by $R_1 >_\succ R_2$, if for any $a \succ b$, $aR_2b \implies aR_1b$.

Corollary 2. (*Comparative Alignments with Preference and Choice Quality*)

Let two DMs have the same $\succ, N$ but with different R_1, R_2 . Suppose R_1 is more aligned with $\succ$ than R_2 , then for any menu A and time t , $\rho^1(\cdot, A, t) \succ_F \rho^2(\cdot, A, t)$ w.r.t $\succ$.

When the salience order becomes more aligned with the preference, items are more likely to be dominated than to be impacted so the support of ρ shrinks. It turns out the converse state is also true. Proposition 6 establishes an equivalence between comparative alignments of salience orders w.r.t the preference and the set inclusion relation for the support of ρ .

Proposition 6. (*Comparative Alignments with Preference and Support Inclusion*)

Suppose ρ_1, ρ_2 have gradual attention representations with $(\succ_1, R_1, N_1), (\succ_2, R_2, N_2)$ such that $\succ_1 = \succ_2 = \succ$. Then the following two statements are equivalent.

1. R_1 is more aligned with $\succ$ than R_2 , i.e., $R_1 >_\succ R_2$.
2. For any menu $A \subseteq X$, $\text{supp}_1(A) \subseteq \text{supp}_2(A)$.

Two preferences can be compared in terms of their alignments with a salience order and Proposition 7 gives a similar result as Proposition 6. Proposition 6 and Proposition 7 are useful for studying aggregation properties of the gradual attention model.

Definition 5. (Comparative Alignments with Saliency)

Let $\succ_1, \succ_2, R \in \mathcal{S}$. $\succ_1$ is said to be *more aligned with R* than $\succ_2$, denoted by $\succ_1 \succ_R \succ_2$, if for any aRb , $a \succ_2 b \implies a \succ_1 b$.

Proposition 7. (Comparative Alignments with Saliency and Support Inclusion)

Suppose ρ_1, ρ_2 have gradual attention representations with $(\succ_1, R_1, N_1), (\succ_2, R_2, N_2)$ such that $R_1 = R_2 = R$. Then the following two statements are equivalent.

1. $\succ_1$ is more aligned with R than $\succ_2$, i.e., $\succ_1 \succ_R \succ_2$.
2. For any menu $A \subseteq X$, $\text{supp}_1(A) \subseteq \text{supp}_2(A)$.

4 Without Rich Choice Data

The gradual attention model requires the researcher to observe choice data from different menus as well as the associated decision time. The rich data requirement might be infeasible in many real world applications. In this section, I study the gradual attention model in two cases without rich choice data. In the first case, choice data with time variation are assumed available but all the data come from a single menu so there is no choice variation across menus. I provide necessary and sufficient conditions to characterize a menu specific gradual attention model. In absence of choice variations across menus, the model primitives are partially identified in general. In the second case, choice data across different menus are assumed available but the decision time information is missing so there is no choice variation over time. In this case, I study a static version of the gradual attention model. I show identification of the model primitives in absence of choice variations over time is generally partial, while a mild *regularity* assumption leads to full identification. The static version of the gradual attention model is directly comparable with other choice models in the literature. In particular, the standard deterministic rational choice model is a limiting case of the static gradual attention model. It can be easily verified that the static gradual attention model is a special case of the monotone random attention model by Cattaneo et al. (2020).

4.1 Without Variation across Menus

Fix a menu A and suppose choice data with time are available but only for the menu A . That is, $\rho(\cdot, A, t)$ for all $t \in \mathcal{T}$ are observed. When $|A| = 2$, Proposition 1 gives a complete characterization of the model. When $|A| > 2$, it is not possible to define dominance and impact relations so the identification results in Proposition 1 are not applicable. As shown in Proposition 2, a key feature of the gradual attention model is the DM makes better choices

as time increases. It turns out this feature can be modified to characterize a menu specific gradual attention model. The modification is necessary since $\succ$ is not always identified for all items in A and there might be multiple candidate preferences consistent with the menu specific model.

Proposition 8. (*Menu Specific Gradual Attention Representation*)

Let A be an arbitrary menu. A random choice rule $\{\rho(\cdot, A, t)\}_{t \in \mathcal{T}}$ has a menu specific gradual attention representation $(\succ_A, R_A, N^A)$ if and only if there exists $P \in \mathcal{S}$ such that $\rho(\cdot, A, t) \succ_F \rho(\cdot, A, s)$ w.r.t $P \in \mathcal{S}$ for all $t > s$. Moreover, the relationship between $P, \succ_A, R_A$ is summarized by the following partial identification results.

- *P is perfectly aligned with $\succ_A$ and is perfectly misaligned with R_A over $\text{supp}(A)$*
- *Suppose $a \in \text{supp}(A), b \in A \setminus \text{supp}(A)$, then $aPb, a \succ_A b, aR_Ab$.*

In Proposition 8, the condition “there exists $P \in \mathcal{S}$ such that $\rho(\cdot, A, t) \succ_F \rho(\cdot, A, s)$ w.r.t $P \in \mathcal{S}$ for all $t > s$ ” is equivalent to saying the DM’s expected utility is increasing over time, regardless of her risk attitudes. This interpretation of the condition is behaviorally more appealing and is more aligned with the increasing choice quality feature of the gradual attention model. In the section of experimental evidence, it is not feasible to exploit choice variations across all possible menus and I focus on testing the menu specific gradual attention model. Thanks to the design of the experiment, the preference $\succ$ is known. It follows from Proposition 8 that the only testable implication is increasing choice quality.

4.2 Without Variations over Time

Suppose the decision time information is not available but choices from all menus are observed. I study a static version of the gradual attention model when there are no choice variations over time. For simplicity, it is assumed the unobservable decision time for all choices are the same. In other words, this section considers a random choice rule ρ with a fixed time index t , where the time dimension of ρ is ignored to save notations. I show even for this simple case, identifying the model primitives is not always possible.

The static version of the gradual attention model studies the same type of choice data as other choice models in the literature. In a standard choice model with limited attention, the model primitives include a preference relation and a probability measure over the set of possible consideration sets for each menu. Choices are modeled as if the DM chooses her most preferred option in her consideration set, where each consideration set is realized according to the probability measure. Unlike these models, the gradual attention model

adds more details to the formation process of the consideration set and uses limited time as a natural justification for limited attention.

The primitives of the static version model are the same as before, except the counting process is replaced by a single probability distribution μ over $\mathbb{Z}^{++}$ determining the size of the consideration set. To make the model more tractable, μ is assumed to have full support on $|X|$, that is, $\mu(Z) > 0$ for all $Z \leq |X|$. Formally, the primitives are $(\succ, R, \mu)$ and ρ is said to have a gradual attention representation without time variation if for all a, A ,

$$\rho(a, A) = \sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ}(A^R(Z)) = a\}} \mu(Z)$$

I claim full identification of $(\succ, R, \mu)$ is generally impossible. The easiest way to see this is to consider a case when X contains only two items a, b and $\rho(a, X) \in (0, 1)$. It can be easily verified the following two different sets of primitives both generate ρ .

$$\begin{aligned} a \succ b \quad bRa \quad \mu(Z = 1) &= \rho(a, \{a, b\}) \\ b \succ' a \quad aR'b \quad \mu'(Z = 1) &= 1 - \rho(a, \{a, b\}) \end{aligned}$$

In contrast, when the decision time information is available, Proposition 1 shows choice variations over time leads to straightforward identification of $\succ, R$ from doubleton menus. It is obvious from the above example, when $|X| = 2$, $\succ, R$ are identified if and only if choices are deterministic. I next show necessary and sufficient conditions for identifying $\succ, R$ when $|X| = 3$, which sheds lights on identification of model primitives in general.

Lemma 2. (*Identification of $\succ, R$ when $|X| = 3$*)

Suppose $|X| = 3$ and ρ has a gradual attention representation without time variation. Then $\succ, R$ can be uniquely identified if and only if either of the following two conditions hold:

1. $\rho(\cdot, A) \in \{0, 1\}$ for all $A \subset X$ such that $|A| = 2$.
2. Positivity holds, that is, $\rho(a, A) > 0$ for all a, A , and $\rho(\cdot, \{a, b\}) \neq \frac{1}{2}$ for all $a, b \in X$.

When the first condition is met, identifying $\succ, R$ is straightforward since the chosen item is more preferred and more salient than the other items. To understand the sufficiency of the second condition, note that positivity implies $\text{supp}(X) = X$, which further implies $\succ, R$ are perfectly misaligned. It follows the choice probabilities from any doubleton menu are $\mu(1), \mu(> 1)$ and choice probabilities from the grand menu X are $\mu(1), \mu(2), \mu(> 2)$. From this observation, $\mu(1)$ can be identified immediately. Since $\rho(\cdot, \{a, b\}) \neq \frac{1}{2}$ for all $a, b \in X$, $\mu(> 2) \neq \mu(1)$ and thus $aRb, b \succ a \iff \rho(a, \{a, b\}) = \mu(1)$.

I next discuss the level of identification that can be achieved, when full identification is not possible. It is useful to introduce a formal definition of non identification. When $\succ, R$ are not identified over a pair of items a, b , these two items are called “equivalent”.

Definition 6. (Equivalent Items)

Let ρ have a gradual attention representation without time variation. Two items a, b are said to be *equivalent*, denoted by $a \sim b$, if there exists $\succ_1, R_1$ and $\succ_2, R_2$ such that

1. $(\succ_1, R_1, \mu_1)$ and $(\succ_2, R_2, \mu_2)$ both generate ρ for some μ_1, μ_2 ;
2. $a \succ_1 b, b R_1 a$ and $b \succ_2 a, a R_2 b$.

To see $\sim$ is indeed an equivalence relation, define $c \sim c$ for all $c \in X$. For any $a \neq b \neq c$, Lemma 3 says it is impossible to have both $a \sim b$ and $b \sim c$. It follows $\sim$ is transitive and thus $\sim$ is an equivalence relation.

Lemma 3. (*Equivalence Class*)

$\sim$ is an equivalence relation and for any item $a \in X$, its equivalence class, denoted by $[a]$, contains at most two elements.

Lemma 3 implies X can be uniquely partitioned into a collection of equivalent classes according to $\sim$. It is straightforward to verify that the restriction of $\succ$ (similarly for R) over the equivalent classes is well defined: $[a] \succ [b]$ if and only if $c \succ d$ for any $c \in [a], d \in [b]$. Proposition 9 shows identification is achieved on the equivalent classes.

Proposition 9. (*Partial Identification*)

Suppose ρ has a gradual attention representation without time variation, then $\succ, R$ are identified up to the equivalence classes. That is, $\sim$ partitions X into a unique collection of equivalence classes and the restriction of $\succ, R$ on the collection of equivalence classes is unique. Moreover, each equivalence class contains at most two items.

The partial identification result in Proposition 9 highlights the novelty of utilizing the choice variations over time in the gradual attention model, where full identification of $\succ, R$ is achieved. The remaining of this section presents results that explore the set of model primitives $\succ, R, \mu$, whose generated random choice rules lead to full identification. One implication of Lemma 2 is that for a general X , $\max_{A \subseteq X} |\text{supp}(A)| \geq 3$ and $\rho(\cdot, \{a, b\}) \neq \frac{1}{2}$ for all $a, b \in X$ together are sufficient for identifying $\succ, R, \mu$. Proposition 10 shows when $\max_{A \subseteq X} |\text{supp}(A)| \geq 3$, identification is generically feasible, in the sense that the set of μ s disabling identification has Lebesgue measure zero.

Proposition 10. (*Generic Conditional Identification*)

Suppose ρ has a gradual attention representation without time variation, and $\max_{A \subseteq X} |\text{supp}(A)| \geq 3$. Then $\succ, R$ are generically identified, that is, the set of μ s disabling identification has Lebesgue measure zero.

Lemma 4 shows the condition $\max_A |\text{supp}(A)| \geq 3$ in Proposition 10 is likely to be satisfied when $|X|$ is large enough, assuming $\succ, R$ are (independently) randomly drawn from a uniform distribution over $\mathcal{S}$.

Lemma 4. (*Asymptotic Support Condition*)

As $|X|$ goes to infinity, the probability that the pair of $\succ, R$ satisfying the $\max_A |\text{supp}(A)| \geq 3$ converges to 1.

5 Aggregation under Heterogeneity

In practice, researchers often do not have access to individual-level choice data and only aggregated choices such as the market share of vehicles are available. This section studies aggregated choice behaviors of a population of heterogeneous agents, where each individual agent's choice is modeled by the gradual attention model. I allow heterogeneity in each of the three model primitives $(\succ, R, N)$, assuming homogeneity in the other two primitives. The main question of interest is whether there exists a representative agent for the population, whose choices are consistent with aggregated choices from the population. The gradual attention model is considered robust to aggregation if a representative exists and the robustness is desirable since it extends the model's applicability.

Formally, I model heterogeneity as a probability distribution over the space of each model primitive. That is, $m^P \in \Delta\mathcal{S}$, $m^S \in \Delta\mathcal{S}$ and $m^C \in \Delta\mathcal{N}$ ⁵ are used for modeling heterogeneity in preferences, salience orders, and counting processes, respectively. The support of m^P is defined by $\text{supp}(m^P) =: \{\succ \in \mathcal{S} : m^P(\succ) > 0\}$ and $\text{supp}(m^S), \text{supp}(m^C)$ are similarly defined⁶. When a representative agent exists, the representative primitives are denoted by $\bar{\succ}, \bar{R}, \bar{N}$ (and its induced $\bar{\mu}$).

Throughout this section, I focus on the gradual attention model with a menu dependent counting process, which is characterized in Theorem 1. Although additional restrictions can be introduced to obtain menu independence as in Proposition 4, the menu dependence of μ is more plausible in reality and adds more flexibility to the gradual attention model. I first show

⁵For a generic measurable set Y , ΔY denotes the set of probability measures over Y with finite support.

⁶Although the same notation $\text{supp}()$ is used, it should not be confused with the support of a menu for a random choice rule.

a representative agent exists under heterogeneity of counting process. Under heterogeneity of salience order or under heterogeneity of preference, I show a representative agent does not exist in general. I present necessary and sufficient conditions for a representative agent to exist under heterogeneity of salience order and under heterogeneity of preference. In particular, I show the choice pattern of increasing quality is robust to aggregation under heterogeneity of counting process and salience order, in a menu specific gradual attention model. Therefore in the section of experimental evidence when aggregated choices are used to test increasing choice quality, assuming experiment subjects to have the same evaluation speed and evaluation order is not necessary.

5.1 Heterogeneity of Counting Process

I start by considering heterogeneity in counting processes so the aggregated choice rule ρ is determined by $\succ, R, m^C$. For each $N \in \text{supp}(m^C)$, the random choice rule generated by $(\succ, R, N)$ is denoted by ρ_N . A representative agent exists if there exist $\bar{\succ}, \bar{R}, \bar{N}$ such that $\rho = \int \rho_N dm^C(N)$ has a gradual attention representation $(\bar{\succ}, \bar{R}, \bar{N})$. Let μ_t^N be probability measure over $\mathbb{Z}^{++}$ that is induced by $N \in \text{supp}(m^C)$. If the representative agent exists, the counting processes must be aggregated as follows: for each $t \in \mathcal{T}, Z \in \mathbb{Z}^{++}$, the probability that the representative agent's consideration set has size equal to Z is $\bar{\mu}_t(Z) = \int \mu_t^N(Z) dm^C(N)$. The following Lemma 5 confirms $\bar{\mu}_t$ is indeed induced by a counting process.

Lemma 5. (*Aggregation of Counting Processes*)

There exists a counting process $\bar{N} \in \mathcal{N}$ that induces $\bar{\mu}_t$.

Proposition 11 says a representative agent exists under heterogeneity of counting process. Its proof comes immediately from Lemma 5 and is omitted. It is not surprising the representative agent shares the same preference and salience order as the population, since homogeneity of $\succ, R$ is assumed. It follows all identification results of $\succ, R$ are unaffected under heterogeneity of counting process and Axioms 1-5 also characterize the aggregated model. In particular, Proposition 11 implies the choice pattern of increasing quality is preserved when experiment subjects have different evaluation speeds in the section of experimental evidence.

Proposition 11. (*Representative Agent under Heterogeneous Counting Processes*)

Suppose ρ is generated by heterogeneous agents with $\succ, R, m^C$, then there exists a representative agent $\bar{\succ} = \succ, \bar{R} = R, \bar{N}$.

5.2 Heterogeneity of Saliency Order

I next consider heterogeneity of saliency orders so the aggregated choice rule ρ is determined by $\succ, m^S, N$. For each $R \in \text{supp}(m^S)$, the random choice rule generated by $(\succ, R, N)$ is denoted by ρ_R . A representative agent exists if there exist $\bar{\succ}, \bar{R}, \bar{N}$ such that $\rho = \int \rho_R dm^S(R)$ has a gradual attention representation $(\bar{\succ}, \bar{R}, \bar{N})$. The following example shows in general, a representative agent does not always exist under heterogeneity of saliency orders.

Example 1. Let $a \succ c \succ b$, aR_1bR_1c , cR_2aR_2b , and $\{R_1, R_2\} = \text{supp}(m^S)$.

To see why, suppose a representative agent exists, then it must be the case $\bar{\succ} = \succ$. Consider the support of each doubleton menu. Note that $\text{supp}(\{a, c\}) = \{a, c\}$ because $a \succ c, cR_2a$, $\text{supp}(\{b, c\}) = \{b, c\}$ because $c \succ b, bR_1c$, and $\text{supp}(\{a, b\}) = a$ because $a \succ b, aR_1b, aR_2b$. It follows $c\bar{R}a\bar{R}b\bar{R}c$, a contradiction to $\bar{R} \in \mathcal{S}$.

Recall that salient orders can be partially ordered by $>_\succ$, in terms of their alignments w.r.t a given preference $\succ$. In the above example, R_1, R_2 cannot be compared by $>_\succ$, which causes inconsistency in the support of doubleton menus. It turns out a necessary and sufficient condition for a representative agent to exist is that the elements of $\text{supp}(m^S)$ can be totally ordered by $>_\succ$ ⁷. When this condition is satisfied, $\bar{R} \in \text{supp}(m^S)$ the minimal element of $\text{supp}(m^S)$.

Proposition 12. (*Gradual Attention with Saliency Heterogeneity*)

Suppose ρ is generated by the gradual attention model with saliency heterogeneity $(\succ, m^S, N)$, then there exists a representative agent with $(\bar{\succ}, \bar{R}, \bar{N})$ if and only if the elements of $\text{supp}(m^S)$ can be totally ordered by $>_\succ$. Moreover, $\bar{R}$ is the minimal element of $\text{supp}(m^S)$, that is, $\bar{R} \in \text{supp}(m^S)$ and $R >_\succ \bar{R}$ for any $R \in \text{supp}(m^S)$.

In Example 1, the inconsistency is generated across the support of doubleton menus. In particular, Axiom 4 is violated. When all choices come from a single menu, Proposition 8 shows Axiom 4 is vacuously true and the only testable implication of the menu specific gradual attention model is increasing choice quality. The following Corollary 3 shows a representative agent always exists in a menu specific gradual attention model under heterogeneity of saliency order. In the section of experimental evidence, aggregated choices are observed within each menu and it is reasonable to believe experiment subjects have different orders of evaluation. Corollary 3 implies the feature of increasing choice quality in a menu specific gradual attention model is robust to aggregation under saliency heterogeneity.

Corollary 3. (*Menu Specific Gradual Attention with Saliency Heterogeneity*)

Fix a menu A . Suppose $\{\rho(\cdot, A, t)\}_{t \in \mathcal{T}}$ is generated by a menu specific gradual attention

⁷It means any two elements of $\text{supp}(m^S)$ can be compared by $>_\succ$.

model with salience heterogeneity $\succ, m^S, N$, then there exists a representative agent with $\succ = \succ, \bar{R}, \bar{N}$.

5.3 Heterogeneity of Preference

Finally, I consider heterogeneity of preferences so the aggregated choice rule ρ is determined by m^P, R, N . For each $\succ \in \text{supp}(m^P)$, the random choice rule generated by $(\succ, R, N)$ is denoted by ρ_R . A representative agent exists if there exist $\bar{\succ}, \bar{R}, \bar{N}$ such that $\rho = \int \rho_{\succ} dm^P$ has a gradual attention representation $(\bar{\succ}, \bar{R}, \bar{N})$. A similar example like Example 1 can be used to show a representative agent does not always exist. Proposition 13 gives a necessary and condition for the existence of a representative agent under preference heterogeneity, which is an analogue to Proposition 12.

Proposition 13. *(Gradual Attention with Preference Heterogeneity)*

Suppose ρ is generated by the gradual attention model with salience heterogeneity (m^P, R, N) , then there exists a representative agent with $(\bar{\succ}, R, \bar{N})$ if and only if the elements of $\text{supp}(m^P)$ can be totally ordered by $>_R$. Moreover, $\bar{\succ}$ is the minimal element of $\text{supp}(m^P)$, that is, $\bar{\succ} \in \text{supp}(m^P)$ and $\succ >_R \bar{\succ}$ for any $\succ \in \text{supp}(m^P)$.

Corollary 4 is a similar result as Corollary 3, which implies the increasing choice quality feature is also robust to aggregation under preference heterogeneity. This result should be interpreted with caution since the aggregated choice quality is defined by the representative agent's preference $\bar{\succ}$. For each individual agent, the expected utility of the aggregated choice probability distribution is not necessarily increasing as time increases.

Corollary 4. *(Menu Specific Gradual Attention with Preference Heterogeneity)*

Fix an arbitrary menu A . Suppose $\{\rho(\cdot, A, t)\}_{t \in \mathcal{T}}$ is generated by a menu dependent gradual attention model with preference heterogeneity m^P, R, N , then there exists a representative agent with $\bar{\succ}, \bar{R} = R, \bar{N}$.

6 Experimental Evidence

In this section, I revisit the experiments by Caplin, Dean, and Martin (2011) and use their data to test the main implication of the gradual attention model: increasing choice quality. Overall I find evidence in support of the hypothesis that experiment subjects make better choices as decision time goes up.

Round
2 of 30

Current selection:

four plus eight minus four

Choose one:

☐	zero	0
☐	three plus five minus seven	1
☐	four plus two plus zero	6
☐	four plus three minus six	1
☒	four plus eight minus four	8
☐	three minus three plus one	0
☐	five plus one minus one	5
☐	eight plus two minus five	5
☐	three plus six minus five	4
☐	four minus two minus one	1
☐	five plus five minus one	9

Finished

Figure 1: An example round (Figure 1 from Caplin, Dean, and Martin (2011))

6.1 Experiment Description

Caplin, Dean, and Martin (2011) conduct four types of experiments and I focus on their “Experiment 2”, which contains the decision time information. In each round of this experiment, subjects need to make a choice from a list of dollar amounts expressed as addition and subtraction operations (see Figure 1 for an example). Subjects are allowed to change their selection at any time during the experiment. Actualized choice is recorded at a random time point that is drawn from a publicly known distribution, truncated at 120 seconds. Each subject took multiple rounds and two rounds were randomly selected, where the recorded choices in these two rounds were actualized for payment.

The experimental design has three desirable features for testing the choice pattern of increasing quality in the gradual attention model. First, decision time is exogenous. Second, the payment mechanism provides incentives for subjects to update their selection instantly when they find a better option, which results in more choice variations over time. Third, the preference is naturally identified since choice objects are dollar amounts. The data collected by Caplin, Dean, and Martin (2011), called “choice process data”, contains the entire choice switching history for each subject in each round. Within a round, it can be shown a subject makes better choices over time if and only if the subject never switches to a worse option. Caplin, Dean, and Martin (2011) find subjects typically switch from lower to higher value objects.

In contrast to a lab environment, it is often not possible to observe an individual’s

entire choice switching history in reality. For example, in a grocery shopping scenario, only customers' final choices at checkout are observed. To closely mimic data feasibility in the real world, I consider aggregating choice data from a population of experiment subjects for each menu at different time points. For each subject in each round, the raw data used for testing only contains information about the randomly selected time point and the subject's choice at the selected time point. I end up with 13902 observations of individual level choice data, after 670 being dropped for practice rounds. I next aggregate the individual level choice data for each menu ⁸. There are 178 distinct menus and the number of observations included in each menu ranges from 29 to 156, with a median equal to 72. Due to the scarcity of data, I do not use the original time in seconds but construct two time bins such that around half of the observations fall into each time bin. Using coarser time bins is equivalent to relabelling time indexes in $\mathcal{T}$ and is thus not affecting any implication of the gradual attention model. Proposition 11 and Corollary 3 show, within a menu, the choice pattern of increasing quality is robust to aggregation under heterogeneity in counting processes and salience orders. Moreover, subjects must have the same preference because of the experiment design. It follows if each subject's choice behavior can be described by the gradual attention model, the aggregated choice of the high time bin must be better than that of the low time bin for each menu.

6.2 Testing Increasing Choice Quality

For each menu, choices in each time bin is aggregated as a probability distribution over the menu of options. For menu i , let F_i^H and F_i^L be empirical cumulative distribution functions (CDF) for the high time bin and the low time bin, respectively. For menu i , testing increasing choice quality is equivalent to testing $F_i^H(x) \leq F_i^L(x)$ for all x in the support of menu i , that is F_i^H FOSD F_i^L . I check $F_i^H(x) \leq F_i^L(x)$ for each menu i and find this condition is satisfied in 122 out of 178 menus. Figure 2 gives two example menus with/without increasing choice quality.

I next conduct statistical tests for FOSD. For menu i , the null hypothesis H_0 its alternative hypothesis H_1 are given as follows.

$$\begin{aligned} H_0 &: F_i^H(x) \leq F_i^L(x) \text{ for all } x \\ H_1 &: F_i^H(x) > F_i^L(x) \text{ for some } x \end{aligned}$$

I adopt the FOSD test proposed by Barrett and Donald (2003). The Barrett-Donald

⁸Each menu is identified by a vector of non-negative integers, which are the list of dollar amounts.

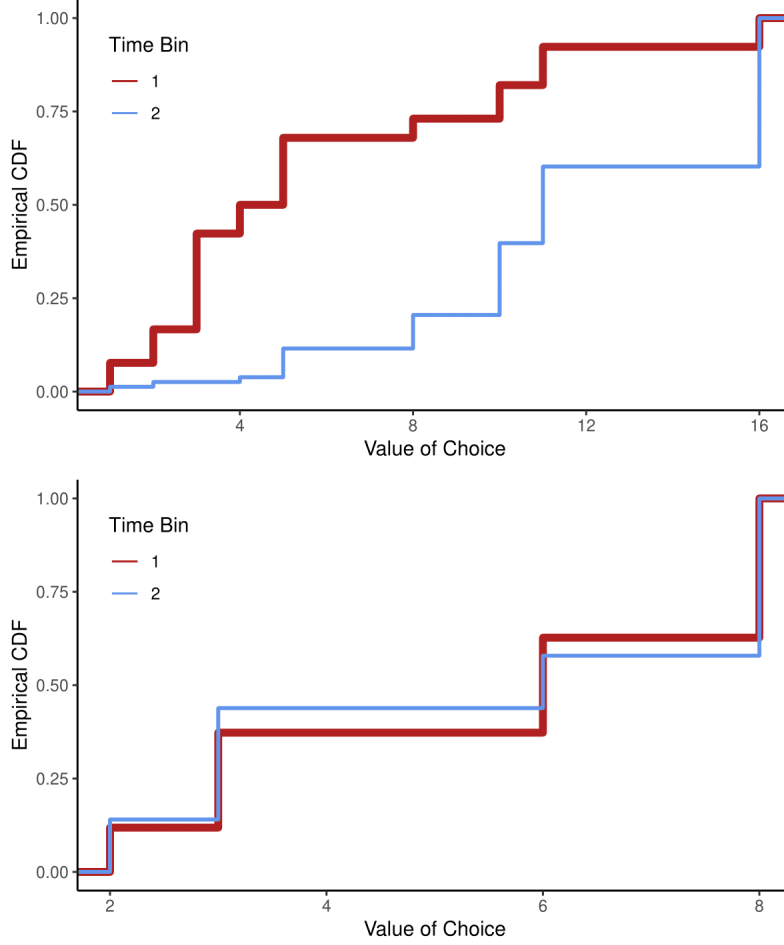


Figure 2: Menus with/without increasing choice quality

(BD) test is a supremum-type test and its test statistic is given by

$$\sqrt{\frac{N_i^H N_i^L}{N_i^H + N_i^L}} \cdot \max_x (F_i^H(x) - F_i^L(x))$$

, where N_i^H, N_i^L are sample sizes for the high time bin and the low time bin in menu i . The confidence interval of the test statistics is computed by bootstrap under the least favorable case of the null hypothesis H_0 , that is, when $F_i^H = F_i^L$. The p-values of the test statistics for all menus are summarized in Figure 3. The red vertical line indicates the 10% critical level. Even at the critical level of 10%, the null hypothesis is rejected in only 5 out of 178 menus. There is only one menu for which the null hypothesis is rejected at 1% level. It is reasonable to believe these rejections are purely due to sampling errors.

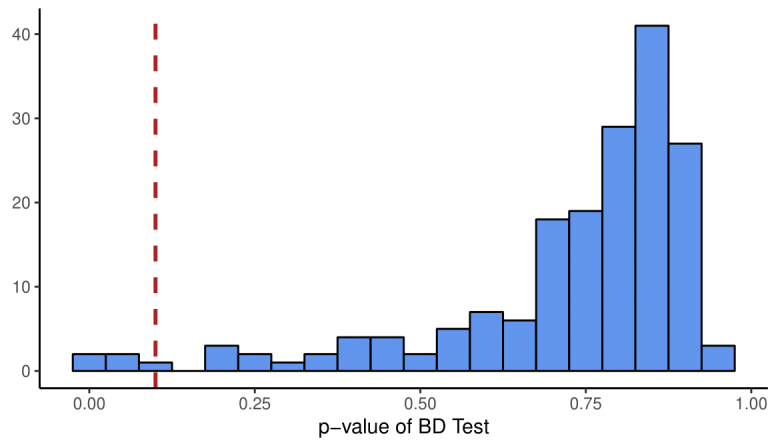


Figure 3: The distribution of p-values from the Barrett-Donald tests

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7 Appendix

7.1 Counting Process and FOSD

A counting process $\{N_t\}_{t \in \mathcal{T}}$ is a non-decreasing stochastic process that takes values in $\mathbb{Z}^+$, that is, for $t \geq s$, $Pr(N_t < N_s | N_s) = 0$. For a counting process $\{N_t\}_{t \in \mathcal{T}}$, its induced probability distribution over $\mathbb{Z}^+$ at time t is denoted by μ_t , $\mu_t(Z) := Pr(N_t = Z)$. The following lemma follows immediately from the definition of a counting process.

Lemma 6. (*Counting Process implies FOSD*)

Suppose the collection of $\{\mu_t\}_{t \in \mathcal{T}}$ is induced by a counting process $\{N_t\}_{t \in \mathcal{T}}$, then $\mu_t \succ_F \mu_s$ w.r.t the natural order $>$ over $\mathbb{Z}^+$ for all $t \geq s$.

Proof. Let $t \geq s$. For any $K \in \mathbb{Z}^+$,

$$\begin{aligned} \sum_{j=0}^K \mu_t(j) &= \sum_{j=0}^K \sum_{i=0}^j [Pr(N_t = j | N_s = i) \cdot Pr(N_s = i)] \text{ by non-decreasing of } N_t \\ &= \sum_{i=0}^K Pr(N_s = i) \cdot \left(\sum_{j=i}^K Pr(N_t = j | N_s = i) \right) \\ &\leq \sum_{i=0}^K Pr(N_s = i) = \sum_{i=0}^K \mu_s(i) \end{aligned}$$

□

The converse statement of Lemma 6 is also true, though there might be multiple different counting processes inducing the same μ as shown in Example 2.

Lemma 7. (*FOSD Characterization of the Counting Process*)

Let $\{\mu_t\}_{t \in \mathcal{T}}$ be a collection of probability distributions over $\mathbb{Z}^+$. Suppose $\mu_t \succ_F \mu_s$ w.r.t the natural order $>$ on $\mathbb{Z}^+$ for all $t \geq s$, then there exists a counting process $\{N_t\}_{t \in \mathcal{T}}$ such that it induces μ_t for all t .

Proof. It suffices to show that for any $t \geq s$, there exists random variables N_t, N_s, N_{t-s} distributed over $\mathbb{Z}^+$ such that $Pr(N_t = Z) = \mu_t(Z)$, $Pr(N_s = Z) = \mu_s(Z)$ for all $Z \in \mathbb{Z}^+$ and $N_t = N_s + N_{t-s}$. This is because I can then construct the transition probabilities from N_s to N_t by $Pr(N_t = Z_1 | N_s = Z_2) = Pr(N_{t-s} = Z_1 - Z_2)$ for all $Z_1, Z_2 \in \mathbb{Z}^+$, where $Pr(N_{t-s} < 0) = 0$ and $Pr(N_t < N_s | N_s) = 0$.

Let F_t, F_s to denote the CDFs for N_t, N_s (over $\mathbb{R}$) and note that $\mu_t \succ_F \mu_s$ implies $F_t(Z) := \sum_{j=0}^Z \mu_t(j) \leq F_s(Z) := \sum_{j=0}^Z \mu_s(j)$ for all $Z \in \mathbb{Z}^+$. Let W be a random variable

that has a uniform distribution over $[0, 1]$. Define a function $G_t : [0, 1] \rightarrow \mathbb{Z}^+$ as follows:

$$G_t(w) := \inf\{Z \in \mathbb{Z}^+ : F_t(Z) \geq w\}$$

Since F_t is a right-continuous step function that jumps only on $\mathbb{Z}^+$, $F_t \circ G_t(w) \geq w$ and $G_t \circ F_t(Z) = Z$ for all $w \in [0, 1], Z \in \mathbb{Z}^+$. I next verify that the random variable $G_t(W)$ has the same CDF as F_t . On one hand, the event $\{G_t(W) \leq Z\} \subseteq \{F_t \circ G_t(W) \leq F_t(Z)\}$ by non-decreasing of F_t so $Pr(G_t(W) \leq Z) \leq Pr(F_t \circ G_t(W) \leq F_t(Z)) \leq Pr(W \leq F_t(Z)) = F_t(Z)$. On the other hand, the event $\{W \leq F_t(Z)\} \subseteq \{G_t(W) \leq G_t \circ F_t(Z)\}$ by non-decreasing of G_t so $F_t(Z) = Pr(W \leq F_t(Z)) \leq Pr(G_t(W) \leq G_t \circ F_t(Z)) = Pr(G_t(W) \leq Z)$. It follows $Pr(G_t(W) \leq Z) = F_t(Z)$. The function G_s is similarly defined such that the random variable $G_s(W)$ has the same CDF as F_s . Let $N_t = G_t(W), N_s = G_s(W)$ and construct the random variable $N_{t-s} := G_t(W) - G_s(W)$. Since $F_t \leq F_s$ everywhere, $G_t \geq G_s$ everywhere so $Pr(N_{t-s} \geq 0) = 1$ as desired. $\square$

I provide an example below to show the counting process in Lemma 7 is not necessarily unique. Moreover, the stochastic process that induces the same distribution does not need to be a counting process.

Example 2. (Multiple Counting Processes)

Suppose the support of $\mu_s, \mu_t (t \geq s)$ contains only $\{1, 2, 3\}$. So μ_s, μ_t can be written as the following vectors.

$$\mu_s = (0.3, 0.3, 0.4) \quad \mu_t = (0.1, 0.2, 0.7)$$

It can be easily checked $\mu_t \succ_F \mu_s$ w.r.t the natural order on $\mathbb{Z}$. Consider the following three transition matrices from s to t . It can be checked that $\mu_s M_i = \mu_t$ for all $i = 1, 2, 3$.

$$M_1 = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad M_2 = \begin{bmatrix} \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & 0 & 1 \end{bmatrix} \quad M_3 = \begin{bmatrix} 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & \frac{1}{4} & \frac{3}{4} \end{bmatrix}$$

It is clear that M_1, M_2 represent two different counting processes while M_3 is not a counting process since probability of decreasing is strictly positive.

7.2 Proof of Proposition 1

Proof. The claim that $\Delta_{ab}(t)$ is constant if and only if $\Delta_{ab}(t) = 1$ or -1 follows immediately from the time variation assumption. I first show $\Delta_{ab}(t)$ is a monotone function in t . There are four possible cases of $\succ, R$ over a, b . If $a \succ b, aRb$ or $b \succ a, bRa$, $\Delta_{ab}(t)$ is a constant function of t . This is because the less salient item is also less preferred and thus is never chosen.

If $a \succ b, bRa$, $\Delta_{ab}(t)$ cannot be constant. This is because the second property of the counting process implies there exists some t_1, t_2 such that $\mu_{t_1}^N(1) > 0, \mu_{t_2}^N(\geq 2) > 0$, which implies $\Delta_{ab}(t_1) < 1, \Delta_{ab}(t_2) > -1$. By the time variation assumption, $\Delta_{ab}(t)$ cannot be constant. $\Delta_{ab}(t)$ is increasing in t since the probability that the consideration set being singleton is decreasing in t . It follows the choice probability of the more preferred item is increasing in t . If $b \succ a, aRb$, a symmetric argument shows $\Delta_{ab}(t)$ must be decreasing in t . To conclude, $\Delta_{ab}(t)$ must be monotonic in t in all these four cases. The four $\iff$ statements follow immediately.

The above argument also proves the only if part for the characterization of the gradual attention representation when $|X| = 2$. It remains to prove the if part. If $\Delta_{ab}(t)$ is constant, then any counting process works. If $\Delta_{ab}(t) \uparrow$, let $\mu(1) = 1 - \Delta_{ab}(t) \uparrow, \mu(2) = \Delta_{ab}(t) \uparrow$. By Lemma 7, there exists a counting process that induces μ . The case of $\Delta_{ab}(t) \downarrow$ is proved similarly. $\square$

7.3 Proof of Proposition 2

Proof. First, note that $t \geq s \implies \mu_t^N \succ_F \mu_s^N$ w.r.t the natural order $>$ over $\mathbb{Z}^+$ by Lemma 6. Second, $A^R(Z') \subseteq A^R(Z)$ for all $Z' < Z$ by the construction of the consideration set. Finally, for all $B' \subseteq B$, either $\mathcal{C}_\succ(B) \succ \mathcal{C}_\succ(B')$ or $\mathcal{C}_\succ(B) = \mathcal{C}_\succ(B')$. Combining these facts proves the result since for any $a \in A$,

$$\begin{aligned} \sum_{\{b \in A: a \succ b\}} \rho(b, A, s) &= \sum_{\{b \in A: a \succ b\}} \sum_{\{Z \in \mathbb{Z}^{++}: \mathcal{C}_\succ(A^R(Z)) = b\}} \mu_s^N(Z) \geq \\ &\sum_{\{b \in A: a \succ b\}} \sum_{\{Z \in \mathbb{Z}^{++}: \mathcal{C}_\succ(A^R(Z)) = b\}} \mu_t^N(Z) = \sum_{\{b \in A: a \succ b\}} \rho(b, A, t) \end{aligned}$$

$\square$

7.4 Proof of Corollary 1

Proof. By representation (1), $\rho(\mathcal{C}_{\succ}(A), A, t) \geq \mu_t(\geq |A|)$ for any menu A . Pick a menu A such that $|A| = K$. Note that for any $\epsilon > 0$, there exists s such that $\mu_s(< |A|) < \epsilon \iff \mu_s(\geq |A|) \geq 1 - \epsilon$. The non-decreasing property of the counting process implies that for any $t \geq s$, $\rho(\mathcal{C}_{\succ}(A), A, t) \geq \mu_t(\geq |A|) \geq \mu_s(\geq |A|) \geq 1 - \epsilon$. $\square$

7.5 Proof of Lemma 1

Proof. Suppose a impacts b and b dominates c . If c dominates a , then Axiom 2 implies b dominates a , a contradiction. If c impacts a , then Axiom 3 implies c impacts b , a contradiction. Axiom 1 implies these two cases together with the two cases in the first part of the statement are exhaustive. A similar argument proves the second part of the statement. $\square$

7.6 Proof of Theorem 1

Proof. I first prove the “only if” direction. Axioms 1-3 follow from Proposition 1. Axiom 4 comes immediately from the representation (1). The Axiom 5 follows from Proposition 2 since the set $\overline{M}(a)$ coincides with the upper contour set of a w.r.t the $\succ$.

I prove the “if” direction by construction and the proof contains five steps.

Step 1. (Construct menu independent $\succ, R \in \mathcal{S}$ from ρ for doubleton menus.)

Construct $\succ$ by $a \succ b \iff a$ dominates b or a impacts b (i.e., $\Delta_{ab}(t) \uparrow$ or $\Delta_{ab}(t) = 1$ for all $t \in \mathcal{T}$). Construct R by $aRb \iff a$ dominates b or b impacts a (i.e., $\Delta_{ab}(t) \downarrow$ or $\Delta_{ab}(t) = 1$ for all $t \in \mathcal{T}$). Axiom 1 guarantees the construction. To verify $\succ, R \in \mathcal{S}$, it remains to check transitivity since the completeness and the anti-symmetry follow directly from the construction.

Let $a \succ b, b \succ c$. If a dominates b , b dominates c (or a impacts b , b impacts c), then Axiom 2 (or Axiom 3) implies a dominates c (or a impacts c). It follows $a \succ c$. If a dominates b , b impacts c (or a impacts b , b dominates c), by Lemma 1, a dominates or impacts c so $a \succ c$.

Let aRb, bRc . If a dominates b , b dominates c (or a is impacted by b , b is impacted by c), then Axiom 2 (or Axiom 3) implies a dominates c (or a is impacted by c), it follows aRc . If a dominates b , b is impacted by c . I prove by contradiction. Suppose cRa , so either c dominates a or a impacts c . But c dominates a and a dominates b implies c dominates b by Axiom 2, which is a contradiction to c impacts b . On the other hand, a impacts c and c impacts b together imply a impacts b by Axiom 3, which is a contradiction to a dominates b . I conclude aRc . The other case when a impacts b and b dominates c can be proved similarly.

Step 2. (Verify the support of ρ is compatible with the constructed $\succ, R$.)

For an arbitrary menu A , I want to verify the $\text{supp}(A)$ is compatible with the constructed $\succ, R$, that is, $a \in A \setminus \text{supp}(A) \iff$ there exists some $b \in A$ such that $b \succ a, bRa$. By construction, $b \succ a, bRa$ implies b dominates a . So the support compatibility condition is satisfied by Axiom 4.

Step 3. (Construct a (menu dependent) counting process N .) For an arbitrary menu A , items in $\text{supp}(A)$ (say $|\text{supp}(A)| = J$) can be labelled as $\text{supp}(A) = \{a^1, a^2, \dots, a^J\}$ such that $a^j \succ a^i \iff j > i$. Since the constructed $\succ, R$ satisfy the support compatibility condition, the restriction of $\succ, R$ over $\text{supp}(A)$ must be perfectly misaligned. It follows $a^i Ra^j \iff j > i$.

Define a function ϕ such that for each a^j , $\phi(a^j) := |\{a \in A : aRa^j\}|$. Note that by construction, $\phi(a^1) = 0$. For $i < j$, $\phi(a^j) - \phi(a^i)$ is the number of items that are less salient than a^i , more salient than a^j , and are never chosen. Now for each $t \in \mathcal{T}$, $\mu_t(Z)$ is constructed as follows.

- If $J = 1$. For any $Z \in \mathbb{Z}^{++}$, $1 \leq Z \leq |X|$, let $\mu_t(Z) = 1/|X|$
- If $J \geq 2$. For each $j \in \{1, 2, \dots, J-1\}$, let $\mu_t(Z) = \rho(a^j, A, t)/(\phi(a^{j+1}) - \phi(a^j))$ for all $Z \in \mathbb{Z}^{++}$ such that $\phi(a^j) \leq Z \leq \phi(a^{j+1})$. For $\phi(a^J) < Z \leq |X|$, let $\mu_t(Z) = \rho(a^J, A, t)/(|X| - \phi(a^J))$.

I **claim** μ is constructed such that $\mu_t \succ_F \mu_s$ w.r.t the natural order over $\mathbb{Z}^{++}$ for any $t \geq s$. By Lemma 7, there exists a counting process N that induces $\{\mu_t\}_{t \in \mathcal{T}}$. It can be verified that N must satisfy the two properties on the counting process by the construction of μ_t . The proof for the claim is given below.

Proof. I want to show for any $1 \leq k \leq |X|$, $\sum_{1 \leq Z \leq k} \mu_t(Z) \leq \sum_{1 \leq Z \leq k} \mu_s(Z)$. When $J = 1$, this is trivially true by construction since $\mu_s(Z) = \mu_t(Z) = \frac{1}{|X|}$. Suppose $J \geq 2$. If $\phi(a^J) < k \leq |X|$,

$$\begin{aligned}
\sum_{1 \leq Z \leq k} \mu_t(Z) &\leq \sum_{1 \leq Z \leq k} \mu_s(Z) \iff \\
\sum_{1 \leq Z \leq \phi(a^J)-1} \mu_t(Z) + \sum_{\phi(a^J) \leq Z \leq |X|} \mu_t(Z) &\leq \sum_{1 \leq Z \leq \phi(a^J)-1} \mu_s(Z) + \sum_{\phi(a^J) \leq Z \leq |X|} \mu_s(Z) \iff \\
\sum_{1 \leq i \leq J-1} \rho(a^i, A, t) + \frac{k - \phi(a^J)}{|X| - \phi(a^J)} \rho(a^J, A, t) &\leq \sum_{1 \leq i \leq J-1} \rho(a^i, A, s) + \frac{k - \phi(a^J)}{|X| - \phi(a^J)} \rho(a^J, A, s) \\
&\iff \rho(a^J, A, s) \geq \rho(a^J, A, t)
\end{aligned}$$

By the construction of $\succ$, the set $\overline{M}(a)$ is equal to $\{b \in X : b \succ a\}$, that is, the upper contour set of a w.r.t $\succ$. By Axiom 5, for any $a \in A$, $\sum_{\{c \in A : c \succ a\}} \rho(c, A, t) \geq \sum_{\{c \in A : c \succ a\}} \rho(c, A, s)$.

Note a^J is labelled such that $a^J \succ c$ for all $c \in A, c \neq a^J$. Let $a \in A$ be an item such that $a^J \succ a \succ c$ for all $c \in A, c \neq a, a^J$. The last line follows from Axiom 5.

If $\phi(a^j) \leq k \leq \phi(a^{j+1})$ for some $1 \leq j \leq J-1$,

$$\begin{aligned} \sum_{1 \leq Z \leq k} \mu_t(Z) &\leq \sum_{1 \leq Z \leq k} \mu_s(Z) \iff \\ \sum_{1 \leq Z \leq \phi(a^j)} \mu_t(Z) + \sum_{\phi(a^j) < Z \leq k} \mu_t(Z) &\leq \sum_{1 \leq Z \leq \phi(a^j)} \mu_s(Z) + \sum_{\phi(a^j) < Z \leq k} \mu_s(Z) \iff \\ \sum_{1 \leq i \leq j-1} \rho(a^i, A, t) + \frac{k - \phi(a^j)}{\phi(a^{j+1}) - \phi(a^j)} \rho(a^j, A, t) &\leq \sum_{1 \leq i \leq j-1} \rho(a^i, A, s) + \frac{k - \phi(a^j)}{\phi(a^{j+1}) - \phi(a^j)} \rho(a^j, A, s) \end{aligned}$$

Axiom 5 implies $\sum_{1 \leq i \leq j-1} \rho(a^i, A, t) \leq \sum_{1 \leq i \leq j-1} \rho(a^i, A, s)$ and $\sum_{1 \leq i \leq j} \rho(a^i, A, t) \leq \sum_{1 \leq i \leq j} \rho(a^i, A, s)$. Suppose the last line above is not true, that is,

$$\sum_{1 \leq i \leq j-1} \rho(a^i, A, t) + \frac{k - \phi(a^j)}{\phi(a^{j+1}) - \phi(a^j)} \rho(a^j, A, t) > \sum_{1 \leq i \leq j-1} \rho(a^i, A, s) + \frac{k - \phi(a^j)}{\phi(a^{j+1}) - \phi(a^j)} \rho(a^j, A, s)$$

Then it must be the case that $\rho(a^j, A, t) > \rho(a^j, A, s)$ since $0 \leq \frac{k - \phi(a^j)}{\phi(a^{j+1}) - \phi(a^j)} \leq 1$. But this yields a contradiction to Axiom 5 since $\sum_{1 \leq i \leq j-1} \rho(a^i, A, t) + \rho(a^j, A, t) = \sum_{1 \leq i \leq j} \rho(a^i, A, t) > \sum_{1 \leq i \leq j} \rho(a^i, A, s) = \sum_{1 \leq i \leq j-1} \rho(a^i, A, s) + \rho(a^j, A, s)$. $\square$

Step 4. (Verify ρ is generated by the gradual attention model with the constructed $(\succ, R, N)$.) I next check that ρ is indeed generated by the gradual attention model with the constructed $(\succ, R, N)$. Let A be an arbitrary menu and suppose $a \in A \setminus \text{supp}(A)$. Since the constructed $\succ, R$ satisfy the support compatibility condition, there exists $b \in A$ such that $b \succ a, bRa$. So for any Z , $A^R(Z)$ either contains both a, b or contains only b . In other words, a is never chosen. It follows $\{Z \in \mathbb{Z}^{++} : \mathcal{C}_\succ(A^R(Z)) = a\} = \emptyset$ and therefore $\rho(a, A, t) = 0 = \sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_\succ(A^R(Z)) = a\}} \mu_t(Z)$.

Suppose $a \in \text{supp}(A)$, say $a = a^j$ for some $1 \leq j \leq J$. Then $\{Z \in \mathbb{Z}^{++} : \mathcal{C}_\succ(A^R(Z)) = a\} = \{Z \in \mathbb{Z}^{++} : a^j \in A^R(Z), a^{j+1} \notin A^R(Z)\} = \{\phi(a^j) + 1, \phi(a^j) + 2, \dots, \phi(a^{j+1})\}$ if $j < J$, and is equal to $\{Z \in \mathbb{Z}^{++} : Z \geq \phi(a^J) + 1\}$ if $j = J$. In the former case, $\sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_\succ(A^R(Z)) = a\}} \mu_t(Z) = (\phi(a^{j+1}) - \phi(a^j)) \cdot [\rho(a^j, D, t) / (\phi(a^{j+1}) - \phi(a^j))] = \rho(a^j, D, t)$. In the latter case, $\sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_\succ(A^R(Z)) = a\}} \mu_t(Z) = 0 + (|X| - \phi(a^J)) \cdot [\rho(a^J, D, t) / (|X| - \phi(a^J))] = \rho(a^J, D, t)$.

Step 5. (Check the uniqueness of $\succ, R$.) Suppose there exists another pair of menu independent $\succ', R'$ generating the same ρ . Proposition 1 immediately implies $\succ = \succ', R = R'$. $\square$

7.7 Proof of Proposition 3

Proof. Let $\bar{Z} = \max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\bar{H}(a) \cap A| = |\bar{H}(a^*) \cap A^*|$, where a^*, A^* is a pair that achieves the maximum (this pair is not necessarily unique). Let time $t \in \mathcal{T}$ be arbitrary. Next I show $\mu_t(Z)$ is identified uniquely for each $1 \leq Z \leq \bar{Z}$ and can be any arbitrary values for $Z > \bar{Z}$ (as long as μ is a valid probability distribution).

Note $\bar{Z} = \max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\bar{H}(a) \cap A| = |\bar{H}(a^*) \cap A^*|$ implies there are $\bar{Z}$ items in A^* that are more salient than a^* , and any item in A^* that is less salient than a^* is never chosen. It follows $0 < \rho(a^*, A^*, t) = \sum_{Z > \bar{Z}}^\infty \mu_t(Z)$. I conclude that for any $Z > \bar{Z}$, $\mu_t(Z)$ can be any arbitrary value subject to the constraint $\sum_{Z > \bar{Z}}^\infty \mu_t(Z) = \rho(a^*, A^*, t)$.

It also follows that $\sum_{a \in \bar{H}(a^*) \cap A^*} \rho(a, A^*, t) = 1 - \sum_{Z > \bar{Z}}^\infty \mu_t(Z) = \sum_{1 \leq Z \leq \bar{Z}} \mu_t(Z)$ since ρ adds up to 1. Since $R \in \mathcal{S}$, the elements in $\bar{H}(a^*) \cap A^*$ can be totally ordered by R and thus can be labelled such that $a_1 R a_2 R \dots R a_{\bar{Z}}$. Now consider the menu $A^* \setminus \{a_{\bar{Z}}\}$. It is clear that $a^* \in \text{supp}(A^* \setminus \{a_{\bar{Z}}\})$ and $\bar{Z} - 1 = |\bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}\}| = \max_{a \in \text{supp}(A^* \setminus \{a_{\bar{Z}}\})} |\bar{H}(a) \cap A^* \setminus \{a_{\bar{Z}}\}|$. A similar argument yields $\sum_{a \in \bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}\}} \rho(a, A^* \setminus \{a_{\bar{Z}}\}, t) = \sum_{1 \leq Z \leq \bar{Z} - 1} \mu_t(Z)$. It follows $\mu_t(\bar{Z}) = \sum_{a \in \bar{H}(a^*) \cap A^*} \rho(a, A^*, t) - \sum_{a \in \bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}\}} \rho(a, A^* \setminus \{a_{\bar{Z}}\}, t)$.

I continue this procedure and for each $Z = 1, \dots, \bar{Z} - 1$, each $\mu_t(Z)$ is identified as

$$\begin{aligned} & \sum_{a \in \bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_{Z+1}\}} \rho(a, A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_{Z+1}\}, t) - \\ & \sum_{a \in \bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_Z\}} \rho(a, A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_Z\}, t) \end{aligned}$$

The uniqueness of $\mu_t(Z)$ for $1 \leq Z \leq \bar{Z}$ follows immediately by this construction. $\square$

7.8 Proof of Proposition 4

Proof. It is straightforward to check the “only if” direction and the proof is omitted. I prove the “if” direction. Theorem 1 shows ρ has a gradual attention representation with menu independent $\succ, R$ and a menu dependent counting process under Axioms 1-5. It remains to show a menu independent μ can be constructed under Axiom 6. Following the proof of Proposition 3, let $\bar{Z} = \max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\bar{H}(a) \cap A| = |\bar{H}(a^*) \cap A^*|$. For an arbitrary $t \in \mathcal{T}$, μ_t is constructed as follows.

Let the elements in $\bar{H}(a^*) \cap A^*$ be labelled such that $a_1 R a_2 R \dots R a_{\bar{Z}}$. Define $\mu_t(\bar{Z}) = \sum_{a \in \bar{H}(a^*) \cap A^*} \rho(a, A^*, t) - \sum_{a \in \bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}\}} \rho(a, A^* \setminus \{a_{\bar{Z}}\}, t)$. For each $Z = 1, \dots, \bar{Z} - 1$, define

$\mu_t(Z)$ by

$$\begin{aligned} & \sum_{a \in \overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{Z+1}\}} \rho(a, A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{Z+1}\}, t) - \\ & \sum_{a \in \overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_Z\}} \rho(a, A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_Z\}, t) \end{aligned}$$

I next show the constructed μ_t is compatible with ρ for arbitrary menus. Let a menu B and $b \in \text{supp}(B)$ be arbitrary. Since ρ has a gradual attention representation with a menu dependent counting process, denote ν as its induced probability measure over $\mathbb{Z}^+$ for menu B . If cRb for all $c \in \text{supp}(B) \setminus \{b\}$, or equivalently, $c \in \text{supp}(B) \setminus \{b\} \implies c \in \overline{H}(b) \cap B$, then $\rho(b, B, t) = 1 - \sum_{1 \leq Z \leq |\overline{H}(b) \cap B|} \nu_t(Z) = 1 - \sum_{c \in \overline{H}(b) \cap B} \rho(c, B, t)$ by the representation. I want to show $\sum_{1 \leq Z \leq |\overline{H}(b) \cap B|} \mu_t(Z) = \sum_{1 \leq Z \leq |\overline{H}(b) \cap B|} \nu_t(Z)$. By Axiom 6 and the fact that $|\overline{H}(b) \cap B| = |\overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b) \cap B|+1}\}|$,

$$\begin{aligned} \sum_{c \in \overline{H}(b) \cap B} \rho(c, B, t) &= \sum_{a \in \overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b) \cap B|+1}\}} \rho(a, A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b) \cap B|+1}\}, t) \\ &= \sum_{1 \leq Z \leq |\overline{H}(b) \cap B|} \mu_t(Z) \text{ by the construction of } \mu_t \end{aligned}$$

If there exists $c \in \text{supp}(B)$ and aRc (or equivalently $c \notin \overline{H}(b) \cap B$), then let b^* be the R -maximal element of $\text{supp}(B) \setminus (\overline{H}(b) \cup \{b\})$, that is, $b^* := \arg \max_{\{c: c \in \text{supp}(B), b \in \overline{H}(c)\}} |\overline{H}(c) \cap B|$. Then $\rho(b, B, t) = \sum_{|\overline{H}(b) \cap B| < Z \leq |\overline{H}(b^*) \cap B|} \nu_t(Z) = \sum_{c \in \overline{H}(b^*) \cap B} \rho(c, B, t) - \sum_{c \in \overline{H}(b) \cap B} \rho(c, B, t)$ by the representation. I next show $\sum_{|\overline{H}(b) \cap B| < Z \leq |\overline{H}(b^*) \cap B|} \nu_t(Z) = \sum_{|\overline{H}(b) \cap B| < Z \leq |\overline{H}(b^*) \cap B|} \mu_t(Z)$. By Axiom 6 and the fact that

$$\begin{aligned} |\overline{H}(b) \cap B| &= |\overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b) \cap B|+1}\}| \\ |\overline{H}(b^*) \cap B| &= |\overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b^*) \cap B|+1}\}| \end{aligned}$$

, I conclude

$$\begin{aligned}
& \sum_{c \in \overline{H}(b^*) \cap B} \rho(c, B, t) - \sum_{c \in \overline{H}(b) \cap B} \rho(c, B, t) = \\
& \sum_{a \in \overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b) \cap B|+1}\}} \rho(a, A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b) \cap B|+1}\}, t) - \\
& \sum_{a \in \overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b^*) \cap B|+1}\}} \rho(a, A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b^*) \cap B|+1}\}, t) \\
& = \sum_{|\overline{H}(b) \cap B| < Z \leq |\overline{H}(b^*) \cap B|} \mu_t(Z) \text{ by the construction of } \mu_t
\end{aligned}$$

□

7.9 Proof of Corollary 2

Proof. It is sufficient to show for any realization of $N_t = Z \in \mathbb{Z}^{++}$, $\mathcal{C}_{\succ}(A^{R_1}(Z)) = \mathcal{C}_{\succ}(A^{R_2}(Z))$ or $\mathcal{C}_{\succ}(A^{R_1}(Z)) \succ \mathcal{C}_{\succ}(A^{R_2}(Z))$. I prove by contradiction. Let $\mathcal{C}_{\succ}(A^{R_2}(Z)) = a$ and suppose $a \succ b$ for any $b \in A^{R_1}(Z)$. If $a R_2 b$, then R_1 being more aligned with $\succ$ implies $a R_1 b$, a contradiction to $a \notin A^{R_1}(Z)$. It follows $b R_2 a$ for any $b \in A^{R_1}(Z)$. But then $A^{R_2}(Z)$ must contain $A^{R_1}(Z)$ and a , a contradiction to $|A^{R_1}(Z)| = |A^{R_2}(Z)|$. □

7.10 Proof of Proposition 6

Proof. 1 $\implies$ 2. Let A be an arbitrary menu. It is equivalent to show that $A \setminus \text{supp}_2(A) \subseteq A \setminus \text{supp}_1(A)$. Let $a \in A \setminus \text{supp}_2(A)$ be arbitrary. Then by Axiom 4, there exists $b \in A$ such that $b \succ a, b R_2 a$. Since R_1 is more aligned with $\succ$ than R_2 , it follows $b R_1 a$ and $a \notin \text{supp}_1(A)$. This implies $\text{supp}_1(A) \subseteq \text{supp}_2(A)$.

2 $\implies$ 1. Let $a, b \in X$ be arbitrary and say $a \succ b, a R_2 b$. Consider the doubleton menu $\{a, b\}$. $\text{supp}_2(\{a, b\}) = \{a\}$ since b is dominated by a . Since $\text{supp}_1(\{a, b\}) \subseteq \text{supp}_2(\{a, b\})$ and $\text{supp}_1(\{a, b\})$ is nonempty, it follows $\text{supp}_1(\{a, b\}) = \{a\}$, which implies $a \succ b, a R_1 b$. □

7.11 Proof of Proposition 7

Proof. 1 $\implies$ 2. Let A be an arbitrary menu. It is equivalent to show that $A \setminus \text{supp}_2(A) \subseteq A \setminus \text{supp}_1(A)$. Let $a \in A \setminus \text{supp}_2(A)$ be arbitrary. Then by Axiom 4, there exists $b \in A$ such that $b \succ_2 a, b R a$. Since $\succ_1$ is more aligned with R than $\succ_2$, it follows $b \succ_1 a$ and $a \notin \text{supp}_1(A)$. This implies $\text{supp}_1(A) \subseteq \text{supp}_2(A)$.

2 $\implies$ 1. Let $a, b \in X$ be arbitrary and say $a \succ_2 b, aRb$. Consider the doubleton menu $\{a, b\}$ and $\text{supp}_2(\{a, b\}) = \{a\}$ since b is dominated by a . Since $\text{supp}_1(\{a, b\}) \subseteq \text{supp}_2(\{a, b\})$ and $\text{supp}_1(\{a, b\})$ is nonempty, it follows $\text{supp}_1(\{a, b\}) = \{a\}$, which implies $a \succ_1 b, aRb$. $\square$

7.12 Proof of Proposition 8

Proof. The “only if” direction follows immediately from Proposition 2 with $\succ$ as the desirable P . I prove the “if” direction.

Fix a menu A . Suppose $\rho(\cdot, A, t) \succ_F \rho(\cdot, A, s)$ w.r.t some order $P \in \mathcal{S}$ for all $t > s$. So the items in $\text{supp}(A)$ (say $|\text{supp}(A)| = J$) can be labelled according to P as $\text{supp}(A) = \{a^1, a^2, \dots, a^J\}$ such that $a^j Pa^i$ if $i < j$. It follows $\sum_{i=1}^j \rho(a^i, A, t) \leq \sum_{i=1}^j \rho(a^i, A, s)$ for all $j = 1, 2, \dots, J$ and for all $t \geq s$.

Define the menu specific preference $\succ_A$ over $\text{supp}(A)$ by $a^i \succ_A a^j$ if $i > j$. Given $\succ_A$ over $\text{supp}(A)$, the menu specific salience order R_A over $\text{supp}(A)$ is defined to be perfectly misaligned with $\succ_A$, that is, $aR_A b \iff b \succ_A a$ for all $a, b \in \text{supp}(A)$. $\succ_A$ can be extended from $\text{supp}(A)$ to A as follows: for any $a \in \text{supp}(A), b \in A \setminus \text{supp}(A)$, let $a \succ_A b$; for those $b, c \in A \setminus \text{supp}(A)$, $\succ_A$ over b, c can be extended arbitrarily. Similarly, R_A can be extended from $\text{supp}(A)$ to A by letting $aR_A b$ for any $a \in \text{supp}(A), b \in A \setminus \text{supp}(A)$. For those $b, c \in A \setminus \text{supp}(A)$, R_A over b, c can be extended arbitrarily.

Let $t \in \mathcal{T}$ be arbitrary. To construct the menu specific counting process, consider a probability distribution μ_t over $\mathbb{Z}^{++}$ defined as follows. If $J \geq 2$, let $\mu_t(Z) = \rho(a^Z, A, t)$ for $1 \leq Z \leq J-1$ and $\mu_t(Z) = (1 - \sum_{1 \leq k \leq J-1} \rho(a^k, A, t)) / (|X| - J + 1)$ for $J \leq Z \leq |X|$. If $J = 1$, let $\mu_t(Z) = 1/|X|$ for $1 \leq Z \leq |X|$. Since $\sum_{i=1}^j \rho(a^i, A, t) \leq \sum_{i=1}^j \rho(a^i, A, s)$ for all $j = 1, 2, \dots, J$ and for all $t \geq s$, it follows $\mu_t \succ_F \mu_s$ w.r.t the natural order $>$ over $\mathbb{Z}^{++}$ for all $t \geq s$. By Lemma 7, there exists a counting process N^A that induces $\{\mu_t\}_{t \in \mathcal{T}}$. It is straightforward to check that the constructed N^A satisfies the two assumptions for the counting processes.

It remains to verify that ρ is generated by the menu specific gradual attention model with $(\succ_A, R_A, N^A)$. If $a \in A \setminus \text{supp}(A)$, then there exists $b \in \text{supp}(A)$ such that $b \succ_A a, bR_A a$. It follows for any Z , $A^R(Z)$ either contains both a, b or does not contain a . Thus $\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ_A}(A^R(Z)) = a\} = \emptyset$ and $\rho(a, A, t) = 0 = \sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ_A}(A^R(Z)) = a\}} \mu_t(Z)$. If $a \in \text{supp}(A)$, say $a = a^j$ for some $1 \leq j \leq J$. Note that a^j is chosen when it is in the consideration set and any a^i with $i > j$ is not in the consideration set. It follows $\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ_A}(A^R(Z)) = a\}$ is equal to $\{Z \in \mathbb{Z}^{++} : A^R(Z) = \{a^1, a^2, \dots, a^j\}\} = \{j\}$ if $j < J$, and is equal to $\{Z \in \mathbb{Z}^{++} : Z \geq J\}$ if $j = J$. In either case, $\rho(a, A, t) = \sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ_A}(A^R(Z)) = a\}} \mu_t(Z)$.

The statements about the relationship between $P, \succ_A, R$ follow directly from the repre-

sentation. □

7.13 Proof of Lemma 2

Proof. I first prove the “if” direction. Condition 1 implies choices are deterministic and $\rho(a, A) = 1 \iff a \succ b, aRb$ for all $b \in A$. It follows $\succ, R$ are uniquely identify from all doubleton menus. For condition 2, I claim positivity $\implies \succ, R$ are perfectly misaligned. This is because otherwise there exist two items a, b such that $a \succ b, aRb \implies \rho(b, X) = 0$, a contradiction. Let items in X be labelled such that $a_1 \succ a_2 \succ a_3$ and $a_3Ra_2Ra_1$. Since ρ has a gradual attention representation without time variation, $\rho(a, \{a, b\}) \in \{\mu(1), \mu(\geq 2)\}$ and $\rho(a, X) \in \{\mu(1), \mu(2), \mu(\geq 3)\}$. Because $\rho(\cdot, \{a, b\}) \neq \frac{1}{2}$ for all $a, b \in X$, $\mu(1) \neq \mu(\geq 2)$. Since $\mu(Z) > 0$ for all $Z \in \mathbb{Z}$, $\mu(\geq 3) < \mu(\geq 2)$.

There is one item b with $\rho(b, \{a, b\}) \neq \rho(b, \{b, c\})$. The item b corresponds to a_2 . It is unique because $\rho(a_1, \{a_1, a_2\}) = \mu(\geq 2) = \rho(a_1, \{a_1, a_3\})$, $\rho(a_3, \{a_1, a_3\}) = \mu(1) = \rho(a_3, \{a_2, a_3\})$ and $\mu(1) \neq \mu(\geq 2)$. Now consider adding b to the menu $\{a, c\}$. There is one item c with $\rho(c, \{a, c\}) = \rho(c, \{a, b, c\})$. The item c corresponds to a_3 . This is because $\rho(a_1, \{a_1, a_3\}) = \mu(\geq 2) > \mu(\geq 3) = \rho(a_1, \{a_1, a_2, a_3\})$. The remaining item a corresponds to a_1 so $\succ, R$ over a, b, c are identified.

I next prove the “only if” direction. Suppose both conditions fail. I want to show there exist two pairs of $\succ_1, R_1, \succ_2, R_2$ that both generate the same random choice rule ρ . There are only three possible cases.

Case 1. Positivity holds but $\rho(\cdot, \{a, b\}) = \frac{1}{2}$ for some $a, b \in X$. Since ρ has a gradual attention representation without time variation, $\mu(1) = \mu(\geq 2) = \frac{1}{2}$ and $\rho(\cdot, B) = \frac{1}{2}$ for any doubleton menu B . It also follows there is one item c with $\rho(c, X) = \frac{1}{2}$. The following $(\succ_1, R_1, \mu_1)$ and $(\succ_2, R_2, \mu_2)$ both generate ρ .

$$\begin{aligned} a \succ_1 b \succ_1 c \quad cR_1bR_1a \quad \mu_1(1) &= \frac{1}{2}, \quad \mu_1(\geq 3) = \rho(b, X) \\ b \succ_2 a \succ_2 c \quad cR_1aR_1b \quad \mu_2(1) &= \frac{1}{2}, \quad \mu_2(\geq 3) = \rho(a, X) \end{aligned}$$

Case 2. Positivity fails for X and exactly one doubleton menu. Say $\rho(a, \{a, c\}) = 1$. The following $(\succ_1, R_1, \mu_1)$ and $(\succ_2, R_2, \mu_2)$ both generate ρ .

$$\begin{aligned} a \succ_1 c \succ_1 b \quad bR_1aR_1c \quad \mu_1(\geq 2) &= \rho(b, X) \\ b \succ_2 a \succ_2 c \quad aR_2cR_2b \quad \mu_2(\geq 2) &= \rho(a, X) \end{aligned}$$

Case 3. Positivity fails for X and exactly two doubleton menus. If $\rho(a, X) = 1$, then

it must be the case that $\rho(a, \{a, c\}) = \rho(a, \{a, b\}) = 1, \rho(b, \{b, c\}) \in (0, 1)$. The following $(\succ_1, R_1, \mu_1)$ and $(\succ_2, R_2, \mu_2)$ both generate ρ .

$$\begin{aligned} a \succ_1 b \succ_1 c \quad aR_1cR_1b \quad \mu_1(\geq 2) &= \rho(b, X) \\ a \succ_2 c \succ_2 b \quad aR_2bR_2c \quad \mu_2(\geq 2) &= \rho(c, X) \end{aligned}$$

If $\text{supp}(X) = \{a, b\}$, then it must be the case that $\rho(c, \{a, c\}) = \rho(c, \{b, c\}) = 0, \rho(a, \{a, b\}) \in (0, 1)$. The following $(\succ_1, R_1, \mu_1)$ and $(\succ_2, R_2, \mu_2)$ both generate ρ .

$$\begin{aligned} a \succ_1 b \succ_1 c \quad bR_1aR_1c \quad \mu_1(\geq 2) &= \rho(a, X) \\ b \succ_2 a \succ_2 c \quad aR_2bR_2c \quad \mu_2(\geq 2) &= \rho(b, X) \end{aligned}$$

□

7.14 Proof of Lemma 3

Proof. Define $c \sim c$ for all $c \in X$ so $\sim$ is reflexive. Suppose $a \sim b$ ($a \neq b$). I claim it is impossible to find $c \neq a \neq b$ such that $c \sim b$. To see this, let $X' = \{a, b, c\}$ and consider ρ for subsets of X' . By the definition of $\sim$ and Lemma 2, one of the following three cases must be true.

Case 1. Positivity holds but $\rho(\cdot, A) = \frac{1}{2}$ for some doubleton menu $A \subset X'$. Since ρ has a gradual attention representation without time variation, $\rho(\cdot, B) = \frac{1}{2}$ for all doubleton menus $B \subset X'$. It also follows that there is one item a^* with $\rho(a^*, X') = \frac{1}{2}$, and $a^*Ra', a' \succ a^*$ for all $a' \in X' \setminus a^*$. But $a \sim b \implies a^* \neq a \neq b$ so the only possible candidate for a^* is c , but $b \sim c$ is then impossible since $cRb, b \succ c$.

Case 2. Positivity fails for X' and exactly one doubleton menu $A \subset X'$. Note that A cannot be $\{a, b\}$ because otherwise $a \sim b$ is impossible. A cannot be $\{b, c\}$ for the same reason. The remaining possible case is $A = \{a, c\}$.

Suppose $\rho(a, \{a, c\}) = 1$, then $a \succ c, aRc$. If $b \succ a \succ c$, then positivity of $\{b, c\} \implies aRcRb$. It follows $\succ, R$ are identified, a contradiction to $a \sim b$. Similarly, if $a \succ b \succ c$, then positivity of $\{b, c\} \implies aRcRb$ so $\succ, R$ are identified, a contradiction to $a \sim b$. Finally, if $a \succ c \succ b$, then positivity of $\{a, b\} \implies bRaRc$ and $\succ, R$ are identified, a contradiction to $a \sim b$. The argument for the case $\rho(c, \{a, c\}) = 1$ is similar.

Case 3. Positivity fails for X' and exactly two doubleton menu of X' . But $a \sim b \implies$ positivity fails for $\{a, c\}, \{b, c\}$, which is a contradiction to $b \sim c$.

To conclude, the above argument shows $\sim$ is (vacuously) transitive so $\sim$ is an equivalence relation and each equivalent class contains at most two elements. □

7.15 Proof of Proposition 9

Proof. It remains to prove the restriction of $\succ, R$ on the equivalent classes are unique. Suppose it is not the case, then there exist $\succ_1, R_1$ and $\succ_2, R_2$ such that for some $[a], [c]$ ($a \neq b$), $[a] \succ_1 [c], [c]R_1[a]$ and $[c] \succ_2 [a], [a]R_2[c]$. The other possibilities for $\succ_1, R_1$ and $\succ_2, R_2$ over $[a], [c]$ are ruled out by the definition of $\sim$. If both $[a], [c]$ contain only one element, then $a \sim c \implies [a] = [c]$, a contradiction. Otherwise, at least one equivalent class contains two elements, say $[a] = \{a, b\}$. But this implies $a \sim b \sim c$, a contradiction to $[a]$ containing at most two elements. $\square$

7.16 Proof of Proposition 10

Proof. When $\max_{A \subseteq X} |supp(A)| \geq 3$, full identification is achieved if $\rho(\cdot, \{a, b\}) \neq \frac{1}{2}$ for any a, b by Lemma 2. It remains to show that the set of μ s that fail to satisfy this condition has Lebesgue measure zero. Note that $\rho(\cdot, \{a, b\}) \neq \frac{1}{2}$ is equivalent to $\mu(1) \neq \frac{1}{2}$. Consider μ as an element from the unitary simplex in the vector space of $\mathbb{R}^{|X|}$ and the subset of μ s with $\mu(1) = \frac{1}{2}$ clearly has Lebesgue measure zero. $\square$

7.17 Proof of Lemma 4

Proof. Let $|X| = K$ and suppose $\succ, R$ are (independently) randomly drawn from a uniform distribution over $\mathcal{S}$. Note $\max_{A \subseteq X} |supp(A)| < 3$ is equivalent to saying it is impossible to find three distinct items a, b, c such that $\succ, R$ are perfectly misaligned over a, b, c . Fix a preference $\succ$ and consider the number of salience orders in $\mathcal{S}$ that are perfectly misaligned with $\succ$ over at least three elements. The number of permutations that avoid permutation *pattern* 123 is known as the Catalan number C_K (see Simion and Schmidt (1985) for example).

$$C_K = \frac{1}{K+1} \binom{2K}{K}$$

So the probability that a salience order randomly drawn is perfectly misaligned with $\succ$ over at least three elements is given by

$$1 - \frac{C_K}{K!} \rightarrow 1 \quad \text{as} \quad K \rightarrow \infty$$

$\square$

It follows $Pr(\max_{A \subseteq X} |supp(A)| \geq 3)$ converges to 1 as $K \rightarrow \infty$.

7.18 Proof of Lemma 5

Proof. It is sufficient to show that for any $t > s$, $\bar{\mu}_t \succ_F \bar{\mu}_s$ w.r.t the natural order over $\mathbb{Z}^{++}$ since by Lemma 7, this implies there exists a counting process $\bar{N}$ that induces $\bar{\mu}_t$.

Note that for any $K \in \mathbb{Z}^{++}$,

$$\begin{aligned} \sum_{Z=1}^K \bar{\mu}_t(Z) &= \sum_{Z=1}^K \int \mu_t^N(Z) dm^C = \int \left(\sum_{Z=1}^K \mu_t^N(Z) \right) dm^C \\ &\leq \int \left(\sum_{Z=1}^K \mu_s^N(Z) \right) dm^C = \sum_{Z=1}^K \int \mu_s^N(Z) dm^C = \sum_{Z=1}^K \bar{\mu}_s(Z) \end{aligned}$$

The inequality comes from the fact that each $N \in \text{supp}(m^C)$ is a counting process. It is straightforward to verify that $\bar{N} \in \mathcal{N}$. $\square$

7.19 Proof of Proposition 12

Proof. I first prove if a representative agent exists, $R \succ_{\succ} \bar{R}$ for any element $R \in \text{supp}(m^S)$. Note that an item a is not in the support of a menu containing it if and only if there exists an item b such that $b \succ a, bRa$ for all $R \in \text{supp}(m^S)$ if and only if $b \succ a, b\bar{R}a$. So for an arbitrary menu, its support under $\bar{R}$ is a subset of its support under any $R \in \text{supp}(m^S)$. By Proposition 6, $R \succ_{\succ} \bar{R}$ for any element $R \in \text{supp}(m^S)$.

The “if” direction can be verified by letting $\bar{R}$ be the minimal element of $\text{supp}(m^S)$ when $\text{supp}(m^S)$ can be totally ordered by $\succ_{\succ}$. The “only if” direction is proved by Example 1. $\square$

7.20 Proof of Corollary 3

Proof. By Proposition 8, it is sufficient to prove that $\rho(\cdot, A, t) \succ_F \rho(\cdot, A, s)$ w.r.t $\succ$ for any $t \geq s$. This is true because for any $a \in A$,

$$\begin{aligned} \sum_{\{b \in A: a \succ b\}} \rho(b, A, t) &\leq \sum_{\{b \in A: a \succ b\}} \rho(b, A, s) \iff \\ \sum_{\{b \in A: a \succ b\}} \int \rho_R(b, A, t) dm^S(R) &\leq \sum_{\{b \in A: a \succ b\}} \int \rho_R(b, A, s) dm^S(R) \iff \\ \int \left(\sum_{\{b \in A: a \succ b\}} \rho_R(b, A, t) \right) dm^S(R) &\leq \int \left(\sum_{\{b \in A: a \succ b\}} \rho_R(b, A, s) \right) dm^S(R) \end{aligned}$$

The last line follows from $\sum_{\{b \in A: a \succ b\}} \rho_R(b, A, t) \leq \sum_{\{b \in A: a \succ b\}} \rho_R(b, A, s)$ for any $R \in \mathcal{S}$ in Proposition 2. $\square$

7.21 Proof of Proposition 13

Proof. I first prove if a representative agent exists, $\succ_i \succ_R \bar{\succ}$ for any element $\succ_i \in \text{supp}(m^P)$. Note that an item a is not in the support of a menu containing it if and only if there exists an item b such that $b \succ_i a, bRa$ for all $\succ_i \in \text{supp}(m^P)$ if and only if $b \bar{\succ} a, bRa$. So for an arbitrary menu, its support under $\bar{\succ}$ is a subset of its support under any $\succ_i \in \text{supp}(m^P)$. By Proposition 7, $\succ_i \succ_R \bar{\succ}$ for any element $\succ_i \in \text{supp}(m^P)$.

The “if” direction can be verified by constructing $\bar{\succ}$ in the same way as in the proof of Corollary 4. The “only if” direction can be proved by replacing $\succ, R_1, R_2$ with $R, \succ_1, \succ_2$ in the Example 1. $\square$

7.22 Proof of Corollary 4

Proof. Note that the support of the aggregated ρ is $\text{supp}(A) = \cup_{\succ_i \in \text{supp}(m^P)} \text{supp}_i(A)$. $\bar{\succ} \in \mathcal{S}$ is constructed as follows.

- Let $\bar{\succ}$ be perfectly misaligned with R over $\text{supp}(A)$ and a^* be the most salient item in $\text{supp}(A)$.
- Let $\bar{\succ}$ be perfectly aligned with R over $\text{supp}(A)$ and b^* be the most salient item in $A \setminus \text{supp}(A)$.
- Let $a^* \bar{\succ} b^*$.

I claim the constructed $\bar{\succ}$ agrees with $\succ_i$ over $\text{supp}_i(A)$ for any $\succ_i \in \text{supp}(m^P)$. To see this, let $a \succ_i b, a, b \in \text{supp}_i(A) \subseteq \text{supp}(A)$. Since both items are in the support, bRa . Suppose $b \bar{\succ} a$, then $a \notin \text{supp}(A)$, a contradiction. For each $\succ_i \in \text{supp}(m^P)$, $\rho_i(\cdot, A, t) \succ_F \rho_i(\cdot, A, s)$ w.r.t $\succ_i$ for any $t \geq s$. It follows from the claim that $\rho_i(\cdot, A, t) \succ_F \rho_i(\cdot, A, s)$ also w.r.t $\bar{\succ}$ for any $t \geq s$. That is for any $a \in A$,

$$\begin{aligned} \sum_{\{b \in A: a \bar{\succ} b\}} \rho_i(b, A, t) &\leq \sum_{\{b \in A: a \bar{\succ} b\}} \rho_i(b, A, s) \text{ for all } \succ_i \in \text{supp}(m^P) \implies \\ \int \left(\sum_{\{b \in A: a \bar{\succ} b\}} \rho_i(b, A, t) \right) dm^P(\succ_i) &\leq \int \left(\sum_{\{b \in A: a \bar{\succ} b\}} \rho_i(b, A, s) \right) dm^P(\succ_i) \iff \\ \sum_{\{b \in A: a \bar{\succ} b\}} \int \rho_i(b, A, t) dm^P(\succ_i) &\leq \sum_{\{b \in A: a \bar{\succ} b\}} \int \rho_i(b, A, s) dm^P(\succ_i) \iff \\ \sum_{\{b \in A: a \bar{\succ} b\}} \rho(b, A, t) &\leq \sum_{\{b \in A: a \bar{\succ} b\}} \rho(b, A, s) \end{aligned}$$

By Proposition 8, a representative agent exists. $\square$